

Interventions meet the constraints beneath them

Three rigorous economics papers · two current applied-development studies and one field-experiment classic. The following article pages are edited reading excerpts, not complete replacements for the originals.

LITERATURE PULSE

Across all three papers, policies work only when they meet the **binding constraint**. In Uganda, public contracts raise firm sales but crowd out private customers and do not improve productivity, suggesting that demand without capacity can reshuffle business rather than transform it. In Colombia, well-designed WhatsApp information actually lowers migrants' program take-up because both procedural detail and unsolicited contact create friction. The Sri Lankan classic supplies the positive case: random capital grants generate very high returns precisely where credit is scarce. The broader trend is a move beyond asking whether an intervention 'works' toward identifying which constraint it relaxes, which new friction it creates, and whose behavior sits at the margin.

01 CURRENT / GENERAL INTEREST · STARTS ON PAGE 2 · *Journal of Development Economics* 183 · 2026 · peer reviewed · CC BY 4.0

Government Demand and Firm Growth

Ugandan VAT, corporate-tax, payroll, and procurement records ask whether selling to government helps firms grow. A staggered event study that adjusts for self-selection and treatment timing finds higher total sales after a first public contract, but no gain in value added per worker and a persistent fall in private-market sales; comparable sales to large private buyers do not produce the same crowd-out. The administrative panel and mechanism survey are unusually rich, though government entry is not randomly assigned and payment delays or unmeasured contract quality may still shape the estimates.

02 CURRENT / PERSUASION AND MIGRATION · STARTS ON PAGE 16 · *World Bank Policy Research Working Paper* 11423 · 2026 · CC BY 3.0 IGO

When Digital Nudges Nag: Evidence on Program Take-up from a WhatsApp Intervention

A randomized trial sent one of three informational WhatsApp videos to 1,375 undocumented Venezuelan migrants eligible for Colombia's regularization program. Assignment reduced take-up by 8 percentage points, or 15 percent: procedural content discouraged some viewers, while unsolicited contact reduced participation even among non-viewers, with effects concentrated among people initially near the enrollment margin. The randomization, administrative outcomes, preregistration, balance tests, and attrition checks are strong; the main caveat is external validity beyond this vulnerable population and messaging channel.

03 CLASSIC / DEVELOPMENT FINANCE · STARTS ON PAGE 31 · *Quarterly Journal of Economics* 123(4) · 2008 · classic · World Bank open edition

Returns to Capital in Microenterprises: Evidence from a Field Experiment

Randomly assigned cash and equipment grants of roughly \$100 or \$200 identify the return to capital among about 400 low-capital Sri Lankan microenterprises. Monthly real returns average roughly 5 percent and are higher for owners with greater measured ability and fewer alternative household resources, supporting a missing-credit interpretation more than a missing-insurance one. Attrition is low and the paper checks spillovers, reporting changes, cash-versus-equipment effects, and nonlinearities; generalization is limited to very small firms and the original published estimates were later corrected for coding errors.

Editorial note: references and appendices were omitted where possible; no included article contributes more than 15 text pages. Full citations and source links remain in the catalog. AI-written synopses were checked against the included source pages.



Regular article

Government demand and firm growth[☆]Bernard Hoekman^{a,b}, Marco Sanfilippo^{c,d}, Filippo Santi^e, Rohit Ticku^a^a European University Institute, Italy^b CEPR, United Kingdom^c University of Turin, Italy^d Collegio Carlo Alberto, Italy^e Catholic University of the Sacred Heart, Italy

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ABSTRACT

Public procurement in developing countries accounts for approximately 13% of GDP, offering a significant market opportunity for domestic firms, especially those facing demand constraints. This paper explores the impact of selling to government entities on firm performance in Uganda, leveraging detailed administrative data and an event study methodology that corrects for firm self-selection and heterogeneity in the timing of the treatment. The results reveal that while firms increase their total sales after entering government contracts, they do not see improvements in productivity (measured by value added per worker). Moreover, sales to non-government customers drop sharply. This reallocation effect is persistent. This effect is not observed when firms start to sell to large private-sector buyers. Drawing on an original survey of firms engaged in public procurement, we highlight potential drivers of these dynamics, including capacity constraints and profitability differences between public and private sector sales.

1. Introduction

Government represents an important market for domestic firms in developing countries. In 2018, on average, government purchases of goods and services in low-income economies accounted for 13% of GDP (Bosio and Djankov, 2020). In many countries, public procurement is designed to pursue objectives in addition to obtaining value for money, such as distributional equity or industrial development.¹ More generally, as government expenditures on goods and services are an important component of GDP, public procurement will impact on economic activity (Cox et al., 2024). Firms in developing coun-

tries often find it difficult to increase and sustain growth because of limited demand for their products and constraints impeding product differentiation, technology adoption and the realization of scale economies (Woodruff, 2018; Goldberg and Reed, 2023; Verhoogen, 2023; Bassi et al., 2022). This is especially true in low-income settings, where capacity under-utilization is particularly high (Walker et al., 2024). Access to procurement contracts may help firms obtain external financing needed to expand supply capacity, improve quality and increase productivity.² Another channel for improving performance is that firms may benefit from learning by doing that leads to more efficient use of inputs or adopting new products/techniques in order

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¹ This may be reflected in price preferences or earmarking of certain types of contracts for domestic firms ("buy national" policies) (Day and Merkert, 2021).

² In the context of advanced economies, studies find that obtaining government contracts may facilitate access to credit, especially for enterprises most likely to be financially constrained — i.e. smaller firms (Hebous and Zimmermann, 2021; di Giovanni et al., 2022; Coviello et al., 2022).

to satisfy the terms of a government contract (Carrillo et al., 2023; Ferraz et al., 2015). Such benefits may have knock on effects on subsequent sales to the private sector.³ Government demand can have drawbacks, too. Large demand shocks relative to firms' size can crowd-out sales to the private sector. This is especially true for (financially-constrained) smaller firms with limited production capacity that may have to decrease the amount of output sold to the private sector in order to deliver their government contracts (di Giovanni et al., 2022). Also, there might be differences in the prices paid by the government relative to private buyers that may introduce distortions in the market and that are usually related to features of the procurement process (Bandiera et al., 2009).

In this paper, we study the performance of Ugandan firms that sell to government entities, analyzing the implications for their growth, labor productivity and sales to and purchases from other firms in the economy. We also consider what happens when firms stop selling to the government.

In Uganda, public procurement is of above average importance, accounting for a large share of GDP (Colonnelli et al., 2024). The government has emphasized the role that public procurement can play in fostering private sector development (NPA, 2020; Behuria, 2021). This is reflected in procurement regulations that aim to increase the share of projects allocated to domestic companies, fostering their participation in procurement (PPDA, 2018b,a, 2019).⁴

We exploit administrative micro data provided by the Uganda Revenue Authority on firm-to-firm transactions from value added tax (VAT) declarations. These are matched with employment level information from the pay as you earn (PAYE) dataset, allowing us to assess different measures of firm performance. VAT declarations pertain to *all* of a firm's buyers and suppliers, including sales to (purchases by) government bodies. This permits us to identify whether and when a firm sells goods or services to a government entity and the value of any such transaction. This feature of the data marks an important difference between our analysis and studies that evaluate the effect of participation in public procurement using quasi-experimental settings based on the design of procurement auctions (e.g. Ferraz et al., 2015; Carrillo et al., 2023). Unlike these studies, we can precisely track the period when firms report sales to government entities and exploit differences in the relationship over time with the government as a buyer. Some firms sell to the government for most of the sample period, while others start or cease to sell to the government, allowing us to identify what happens when they "enter" and "exit".⁵ In addition, we cover firms that may not be captured by public procurement auction data, e.g., smaller purchases awarded at the discretion of purchasing authorities. These are likely to be especially relevant to SMEs.

Our baseline specification links indicators of firms' performance to either the value of their sales to the government or to a dummy variable

that switches to one in the year in which they start selling to a public entity. To address potential endogeneity due, for instance, to possible self-selection by firms, we exploit information for a sub-sample of firms that start providing goods or services to the government after 2012. Given the differences between firms that sell to government entities and firms that do not, we construct the control group based only on not yet treated firms (i.e. firms that will start selling to the government in subsequent years). This reduces concerns about potential self-selection of some types of firms into government contracts, allowing for more precise identification of the effect of starting (or stopping) to sell to the public sector. As the distribution of the treatment timing is heterogeneous across firms over time, we exploit recent advances in the two-way fixed effects literature that address potential biases in event study coefficient estimates stemming from the presence of negative weights under parallel trends and no anticipation effect assumptions (De Chaisemartin and D'Haultfoeuille, 2022; Roth et al., 2023). More specifically, we use the doubly robust estimator proposed by Callaway and Sant'Anna (2021) for settings with multiple periods and multiple treatment cohorts (hereafter, CS estimator). The CS estimator combines the outcome regression approach and inverse probability weighting (IPW) using pre-treatment characteristics to either condition parallel trends or weight the control group by the probability of being treated.

We show that selling to government entities leads to increases in total sales but we do not observe an increase in value added per employee. For treated firms, government contracts do not simply add a new revenue stream that expands sales relative to what was realized before. Firms that start selling to the government reduce sales to other buyers both within the period in which sales to the government commence and thereafter. Transaction-level information reveals that this crowding-out of private sales is associated with a shift in the composition of existing buyers, with an increase in their number and a drop in the average size of each transaction. When firms stop selling to government buyers and do not start again we observe a symmetric relationship: total sales drop as firms are unable to fully replace lost procurement contracts with sales to private buyers. This pattern is specific to transactions involving the government. When replicating the analysis using the first contract with either a (i) large domestic buyer or (ii) foreign-owned firm, we find evidence of total sales increasing without an accompanying drop in sales to other private buyers.

Qualitative evidence obtained from an original survey of 236 Ugandan firms with experience with public procurement suggest two potential mechanisms that may drive our main findings: (1) capacity-cum-credit constraints, and (2) differences in the profitability of sales to public vs. private buyers. We draw on the survey findings to undertake additional data analysis. The survey confirms the prevalence of capacity constraints in many firms that may hamper the possibility of expanding production after obtaining government contracts. We document that firms in services sectors, which rely less on fixed assets and thus may have less need to finance investment in plant and equipment in order to expand production, experience a smaller crowding out effect, consistent with potential (short-term) capacity constraints.⁶ However, a significant share (40% of surveyed firms) expanded production capacity and most surveyed firms did not regard access to finance as a significant factor inhibiting growth, suggesting capacity-cum-credit constraints do not uniquely explain our results. A majority of surveyed firms report that relative to private buyers, government entities pay higher prices and that this motivates reductions in sales to other firms. Suggestive evidence for this hypothesis is obtained both by evidence on firms reporting higher profits after selling to the Government, and by using customs clearance (trade transactions) data for firms active in distribution activities (wholesale and retail trade). Firms in this sector rely

³ Lee (2022) finds that Korean firms selling to the Government increase their sales to the private sector because of a reputational effect associated with success in the public procurement market.

⁴ The National Development Plans (NDP I, II and III) make specific reference to the importance of local content provisions in public procurement. The potential to increase the contribution of domestic factors of production in large-scale projects in the construction and energy sectors is stressed, but also emphasized for smaller-scale projects involving the supply of goods or services in the health, education and defence sectors (PPDA, 2019). The Buy Uganda Build Uganda (BUBU) policy, launched in 2014, aims to increase the value of procurement that goes to local firms (with a specific focus on micro and small firms, which account for some 90% of all firms).

⁵ The richness of the bilateral VAT transaction data permits us to identify four different patterns: (1) the majority of firms (43%) reports positive sales to a government entity throughout the sample period (*Regular* suppliers); a quarter of the treated firms start and stop selling at some point during the period covered by the data (*Irregular* suppliers); (3) one fifth keep selling to the government for all periods after the first transaction (*Stayers*); and (4) about 10% of the firms stopped selling to the government on a definitive basis during the period covered by our sample (*Exiters*).

⁶ di Giovanni et al. (2022) also discuss the reallocation of supply by firms. In their model, this is driven by short-run capital constraints which are resolved by procurement contracts acting as collateral.

extensively on imports for their sales. Trade data allow decomposing their imports into quantity and unit prices. Using information on the share of imports in total sales, we show that for this subsample of firms sales to the government are associated with higher prices.

Related research. Our paper speaks to the literature on barriers to firm growth in developing countries. Most extant research has focused on the supply-side (see Woodruff, 2018, for a review). Our contribution is on demand-side constraints (Goldberg and Reed, 2023) and the potential role of sales to the government in alleviating such constraints. Egger et al. (2022) document an increase in sales realized by firms following a positive demand shock (driven by a large, one time cash transfer to local households) without changes in firm investment behavior, i.e., increased sales do not reflect a relaxation of credit constraints that previously limited investment.

Much of the work on the role of government demand is on the design of public procurement processes to attain value for money and public policy objectives (e.g. Bosio et al., 2022) by reducing information barriers and increasing participation (Colonnelli et al., 2024; Hjort et al., 2024). Less attention has been given to the role participation in procurement as such can play as an instrument to promote private sector development. The evidence from developing countries is generally consistent with what we find, i.e. a positive effect of procurement on several dimensions of firms' performance (Ferraz et al., 2015; Hoekman and Sanfilippo, 2020; Fadic, 2020; Lee, 2022; Mensah et al., 2026). Most existing studies in the development literature focus on middle-income countries. We focus on a low-income setting where firms are more likely to exhibit slack (Walker et al., 2024), helping to improve understanding of the nuanced effects of demand shocks associated with winning government contracts. An important dimension we emphasize in this paper is that the relationship with total sales may not be transitory, a result in line with what Ferraz et al. (2015) and Hjort et al. (2024) find for Brazilian and Liberian firms, respectively.

To our knowledge, the observed reallocation of supply by firms that start to sell to government entities has not been a focus of analysis in the literature on the firm-level effects of participation in public procurement. This reallocation effect is a central feature of the seminal paper by Baldwin and Richardson (1972), who focus on the aggregate (economy-wide) impact of public procurement policies that provide preferential treatment for certain types of firms. They argue that in perfectly competitive markets where government demand is small relative to total market demand, directing procurement to preferred suppliers will have no effect on prices and simply serve to reallocate demand between public and private sector. As government demand is relatively large for many sectors in Uganda, procurement policy may have effects on prices.

Our paper provides firm level evidence of the reallocation effect. This marks a difference with the findings by Ferraz et al. (2015) and Carrillo et al. (2023), who find no evidence for sales to private sector being displaced by sales to the government in their samples of Brazilian, Ecuadorean, and Kenyan firms.⁷ (di Giovanni et al., 2022) find a short-run crowding out effect of public procurement on private sector sales in Spain, attributing this to financial constraints that limit production capacity. A similar pattern is documented by Mensah et al. (2026) in Kenya: firms entering substantial sales relationships with government entities experience a contemporaneous crowding out of private-sector sales, followed by a subsequent rebound and growth.

⁷ Ferraz et al. (2015) do not find sales to private sector are displaced by sales to the Government. Rather, they find the two to be complementary. A difference with our setting is that they report that Government contracts are fulfilled in less than 1 year, whereas the average exposure of Ugandan firms to transactions with Government entities in our sample is around 3 years. Carrillo et al. (2023) find no evidence of short-term declines in sales to other buyers for a sample of Ecuadorian firms winning procurement contracts in the construction sector.

Unlike previous research, a key, distinctive feature of our paper is that we distinguish between firms based on how frequently they sell to the government and whether they exit from government contractual relationships. We find that obtaining government contracts leads to a persistent decline in sales to the private sector, that is temporarily compensated by an overall increase in the value of total sales. Our survey results suggests that this decline stems partly from Ugandan firms' preference for selling to the government over serving private-sector buyers.

Our findings parallel those of Alfaro-Urena et al. (2022) for Costa Rican firms that start selling to foreign multinationals, which is accompanied with a drop in sales to other buyers. They attribute the decrease to short-term capacity constraints that increase marginal costs (a similar mechanism to that proposed by Almunia et al., 2021), making prices less attractive to private buyers. The majority of firms in our survey report that the government pays higher prices and that this is mostly relevant for firms with high fixed costs. This survey result is consistent with (Alfaro-Urena et al., 2022)'s model. However, in contrast to their conclusion that the decline in private sector sales is transitory, the Ugandan firms in our sample do not observe a recovery in sales to private buyers in subsequent years. This might imply that firms in Uganda cannot invest in new fixed assets, consistent with evidence from a survey of small firms in Uganda by Bassi et al. (2022), or that they do not have sufficient incentives to do so.

The remainder of the paper proceeds as follows. Section 2 presents the data and descriptive statistics on the extent to which firms in Uganda sell to government entities. Section 3 introduces the empirical framework used for analysis and descriptive evidence on the relationship between selling to government entities and firm performance. Section 4 presents our main results, while Section 5 reports robustness checks. An analysis of potential mechanisms, based on administrative data and a survey of Ugandan firms that participate in procurement, is presented in Section 6. Section 7 concludes.

2. Data

The transactions data used in this paper come from administrative records compiled by the Uganda Revenue Authority (URA) for 2012–2019. We use masked (i.e. pseudo-anonymized) records from the Ugandan VAT registry as the main source of data.⁸ The VAT was introduced in Uganda in 1996, with a basic 18% rate and a set of exemptions. In a country with a tax-to-GDP ratio of just 14%, four percentage points below the Sub-Saharan Africa average,⁹ the VAT accounts for over 30% of national tax revenues (Almunia et al., 2024; Brockmeyer et al., 2024). As of 2012, all registered firms above a certain size threshold were required to provide VAT declarations on a monthly basis using an electronic form.¹⁰ The data therefore provide a representative snapshot of all formal bilateral transactions between registered firms in the country.¹¹

Important given the scope of this work, registered firms are those entitled to supply the Government. Participation in standard competitive bidding typically requires registration as a provider with the PPDA and the electronic government procurement (e-GP) system, with local firms required to provide their Tax Identification Number (TIN) to validate their eligibility and complete their supplier registration on the system.¹² While PPDA provide rules for different procurement

⁸ Each record is a precise VAT return event by a given registered entity, and might involve several distinct invoices and partner entities at once.

⁹ See the UNU-Wider Government Revenue Dataset, accessed in July 2025.

¹⁰ This threshold was in the order of UGX 50 million (some USD 13,700) up to 2015, and UGX 150 million thereafter (Almunia et al., 2017).

¹¹ If compared to GDP at market prices produced by the formal sector, as reported by the Uganda Bureau of Statistics (accessed on January 12, 2026), the share of sales reported by firms in our sample is, on average, 61%.

¹² See the e-GP supplier registration guide, accessed on January 12, 2026.

methods, depending on estimated contract values and circumstances, the standard competitive process requires formal documentation and compliance with regulatory requirements. We therefore expect firms covered by the tax registry to account for the vast majority of total public procurement in Uganda.¹³

We build our firm level dataset starting from the Schedule 1 of the VAT declaration form. Schedule 1 consists of a bilateral firm-to-firm transaction level dataset reporting all invoices registered by a firm while filing for the VAT return, and involving other registered firms as partners.¹⁴

We aggregate data at the level of each reporting firm and by year. In total, we count 18,110 reporting firms, for a total of 73,778 firm-year observations over the period considered. Many firms report sporadically prior to 2012 and coverage increases in more recent years. Over 76% of the firms report information for less than 6 years, slightly more than half for less than 3 years. This translates into an unbalanced panel in which firms are included based on whether they present a VAT declaration in a given year. A potential concern is that firms reporting behavior is influenced by their relationship with the Government. To verify if this kind of attrition is relevant in our sample, we run a specification in which a firm's likelihood of being in the data is related to a binary variable that equals one if the firm sell to the government. Results reported in Table L-1 in Appendix L show evidence of a small positive increase in the probability of reporting that disappears once we control for firm and year fixed effects simultaneously.

After aggregating, we match the VAT data to the registry of taxpayers to obtain basic information on firm characteristics such as the year of incorporation, sector of activity (at the 4 digit level of the ISIC rev. 4 classification), and type of organization.¹⁵ The data distinguish two categories of public bodies, "Government bodies and government-funded project" and "Local Authority". We use this information to identify the year a firm reports sales to an entity classified in one of these categories. While many firms sell to a government entity at some point in the sample period, transactions involving the government as a buyer represent only 4.3% of the total number of VAT transactions recorded over the period considered. The average value of a transaction involving a government entity as a buyer is UGX 100 million (approximately US\$ 27,000), about 3 times larger than the average firm-to-firm transaction (UGX 36 million or US\$ 9700).

We integrate the records from the VAT declarations with employment information from the Pay-As-You-Earn (PAYE) dataset. Matching PAYE with VAT data reduces the size of the sample as the concordance between the two datasets is imperfect. Out of the 18,110 unique firms, only 7936 firms have a matching record in the PAYE database. This

¹³ Uganda's PPDA also provides for simplified or micro procurement methods for low-value purchases (below USD 1000 USD or USD 2000 for works) where formal bids are not solicited through the standard tender process. In principle it is therefore possible for informal firms to apply. However, such transactions are structurally distinct from the competitive procurements that generate comprehensive contract award data. Calculations using [publicly available data on contracts reported by the PPDA](#) indicate that these contracts account for less than 2.4% of total procurement value in recent years.

¹⁴ Overall, the data include 6,496,014 transactions over the period 2012–2019. Of these, 6,279,938 can be assigned to one of the registered firms listed in the tax registry (i.e. an average about 320 transactions per firm). Table A-3 in the Appendix report detailed information on the number of transactions per firm and firms' pair. Note that these data are based on monthly reports, i.e. before we aggregate them up at the firm-year level.

¹⁵ Not all firms in the VAT data match with an entry in the tax registry. More significant is a large number of unmatched entries on the declared partner side. In total, there are 15,150 matched reporting firms in the VAT data (accounting for 83.4% of the aggregated sample from Schedule 1 declarations). This discrepancy is likely due to the registry covering firms established up to 2018 (so that younger ones cannot be matched), as well as the possibility that some tax return forms report the wrong partner or reporter tax identification numbers.

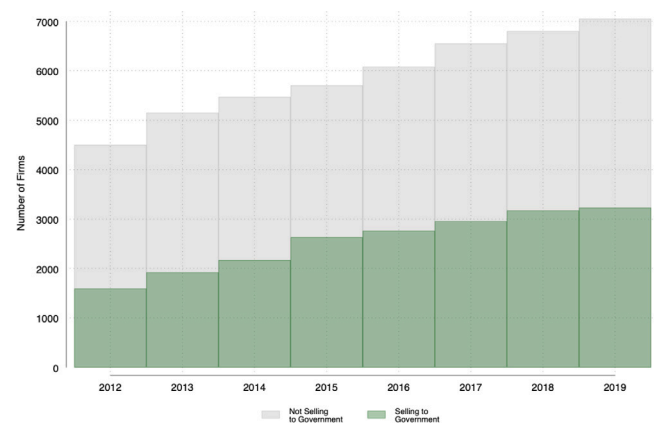


Fig. 1. Government suppliers over time.

Notes: The figure plots the distribution of firms selling and not selling to the government over the period of interest in the full sample.

Source: Authors elaboration on VAT data.

results in 21,749 firm-year pairs. Appendix Tables B-3 and B-4 report the summary statistics for the main outcomes of interest and the distribution of firm-year observations by type of suppliers showing the samples resulting from the different steps of the merge.

2.1. Descriptive evidence on firms selling to the government

About 35% of the firms report at least one transaction with the government over the sample period, with an increase in their number over time (Fig. 1). Appendix Table A-1 provides descriptive statistics on the sectoral distribution of firms with a match in the VAT registry and the share of firms reporting at least one transaction with a government entity.

Compared to other firms, those selling to the government (i) are more likely to be concentrated in distribution services, construction, manufacturing and high value-added services activities (ICT and Technical Professions) (Table A-1); (ii) tend to be larger in terms of sales, a feature that is particularly evident for firms that keep on selling to the government from the time they become a supplier (Table A-2); (iii) have more partners to which they sell and from which they source (Table A-4); and (iv) are older (Figure A-1).

An important feature of the VAT data is that it is possible to track the different patterns of firm interactions with the government entities. Aside from firms that never sell to the government, we can distinguish between four groups of firms:

1. *Regular suppliers*: 43.7% of firms report sales to government bodies throughout the sample period;
2. *Stayers*: 20.2% of firms record a first sale to a government entity after they appear in the sample and continue to report transactions thereafter;
3. *Irregular suppliers*: 25.1% of firms record a transaction with a public body after they first appear in the sample but subsequently engages irregularly with the government during the remainder of the sample period¹⁶;
4. *Exiters*: 11.1% of firms sell to a government entity the first time they appear in the sample, but stop doing so at some point subsequently.

¹⁶ For instance, a firm reports its first sale to the government in 2013, another in 2016, stops in 2017 and starts again in 2018.

Table 1
Summary statistics and group comparisons on selected outcomes.

	Never suppliers		Government suppliers		Mean Diff (G2-G1)	t-stat
	Mean	SD	Mean	SD		
Total sales	1782.65	(18 997.64)	5125.38	(44 509.23)	3342.73***	12.32
Sales Excluding Govt.	1782.65	(18 997.64)	4281.07	(41 972.21)	2498.42***	9.69
Buyers Excluding Govt.	9.55	(23.47)	37.60	(145.61)	28.05***	33.59
Average Sales excl. Govt.	360.42	(3585.07)	540.58	(18 721.40)	180.16*	1.67
Sales per employee	420.94	(1961.67)	408.14	(1733.22)	-12.81	-0.59
Value added per employee	-172.63	(6736.87)	-22.39	(1366.86)	150.23**	2.57

Notes: Standard deviations in parentheses. All outcomes except for *Buyers Excluding Govt.* are expressed in millions UGX. Mean Diff shows the difference between firms that sell to the government at least once over the period of interest and firms that never sell to the government - i.e. the “never treated”. The negative mean values of value added per employee for never-suppliers and government suppliers are driven by outlier firms. The median values for the two groups are 1.38 million UGX and 7.26 million UGX, respectively. The *t*-statistic tests whether the mean difference is significantly different from zero. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Descriptive evidence on the main differences between firms that sell to the government at least once as compared to those that never sell to the government over the period of interest is reported in Table 1. Overall, firms that at some point sell to government entities at least once have larger sales volume, a bigger sales network (as proxied by the average number of buyers excluding government), have larger average gross profits and appear to be relatively more productive than firms that never sell to the government. Additional descriptive statistics on the variables of interest for the whole sample are reported in Appendix Tables B-1 and B-2.

3. Empirical analysis

3.1. Regression analysis

We begin by examining the relationship between selling to the government and firm performance in a standard two-Way fixed effects (TWFE) setting:

$$Y_{it} = \beta_1 Gov_{it} + \lambda_i + \theta_{st} + \varepsilon_{it} \quad (1)$$

where Y_{it} is the outcome of interest for firm i in year t . Specifically, we look at sales to different registered partners using the full sample from the bilateral VAT data. We also look at the domestic network of firms, i.e. the number of buyers and the average size of transactions with private domestic buyers. For the sample of firms that can be matched with PAYE data, we develop measures of labor productivity (sales and value-added per employee).¹⁷

In this specification, the explanatory variable, Gov_{it} , measures either the (log) value of sales to government entities by firm i at time t or a dummy, taking the value 1 when a firm starts selling to the government and zero otherwise. λ_i , and θ_{st} are firm fixed effects and sector (at the 4 digit ISIC classification) time trends, respectively. Firm fixed effects account for firm-specific unobserved factors, while the addition of sector-specific fixed effects accounts for industry-specific factors that may influence firm performance, such as sector-specific trade or industrial policies. Standard errors are clustered at the firm level. Estimates based on Eq. (1) are reported in Appendix Table C-1 for both the discrete definition of the treatment (selling to the government) and a continuous definition of the treatment (value of firm sales to the government). Results are consistent across the two treatment specifications. Selling to government entities is associated with (i) higher levels of total sales and (ii) lower levels of sales to non-government buyers. The latter finding provides some suggestive evidence that sales to government entities come at the expense of sales to other buyers, i.e., involve some reallocation of capacity instead of being additional. We also observe changes in the composition of the group of private buyers after a firm obtains a government contract.

¹⁷ The value-added measure is computed by subtracting purchases from registered partners and imports from sales to registered partners and exports.

While the firm expands its network of private sector buyers, the average size of transactions with private sector buyers reduces. Selling to the government is associated with an improvement in labor productivity, defined as sales per worker, and in value-added per worker. This preliminary set of results are robust to (a) using a sub-sample of firms that is the same across all specifications (Appendix Table C-2); and (b) a sub-sample of firms with a balanced panel (Appendix Table C-3).

3.2. Event study analysis

While including a broad set of fixed effects helps to account for omitted variables, both reverse causation due to self-selection of certain types of firms and measurement error are potential sources of bias that we cannot account for. The results from Eq. (1) therefore cannot be interpreted in a causal manner given that firms that sell to government entities are unlikely to have performed similarly to firms that do not, absent the treatment. Table 2 uses the four-fold categorization discussed previously to compare the characteristics of treated firms before the first transaction with the government with firms that never sell to the government. The differences are substantial for the variables included.¹⁸ There are also relevant differences among firms that sell to the government. Firms in the group of *Stayers* are larger and have higher turnover and assets than those in the irregular supplier group.

In light of this we implement an event study approach to examine if the trajectory of the estimated effect on firm performance changes discretely in response to the first sale to a government buyer.¹⁹ The event study design also enables us to identify if there is a persistent effect of receiving government contracts on firm performance. To rule out any pre-trends in the performance across firms that begin selling to the government and those that never do so, as appears to be the case (Table 2 and Appendix Table A-5), we restrict the control group to firms that *will* sell to the government in the future, that is, the not-yet-treated firms (i.e. firms that will eventually start selling to the government over our period of interest).

Even in the absence of pre-trends, estimates could be contaminated by the treatment effects from other periods (see Roth et al., 2023; De Chaisemartin and D'Haultfoeuille, 2022). As the treatment is heterogeneous over time across different units, the effects we find may not provide the correct weighted average of treatment effects across units (in our case, firms) and time. This is because, with heterogeneous treatment, we end up comparing treated firms with not yet treated ones, which is correct, and with already-treated firms, which is not.

¹⁸ Appendix Table A-5 shows that firms selling to the government tend to have fewer buyers and are more profitable relative to firms that never sell to the government, even before they start selling to the government (Panel b vs. Panel a). Table A-6 reports similar statistics for suppliers using the pattern identified in Section 2.

¹⁹ In the recent FDI literature, event study designs have been used to study the effect of becoming a supplier to multinational corporations in Costa Rica and Belgium (Alfaro-Urena et al., 2022; Amity et al., 2024).

Table 2

Group difference (Before first sale to the government).

	Total sales (UGX mln)	Total assets (UGX mln)	Number of employees	Number of buyers	Number of suppliers
Stayers	5076.85	9083.93	41.50	16.19	18.87
Irregular sellers	1959.96	3438.35	35.30	16.89	18.50
Stayers or Irregular sellers	3563.68	6702.27	38.41	16.53	18.69
Never sellers	1099.40	1966.54	21.96	6.89	10.25
Stayers vs. Never	0	0	.00007	0	0
Irregulars vs. Never	.09232	.08046	.00703	0	0

Notes: Mean values for selected outcomes across groups, defined by the type of economic relationship over the period of interest. Row 4 and 5 report the p-value from a two-tailed Wald test for mean difference across groups for stayers vs. never sellers and intermittent vs. never sellers respectively. Stayers consist of firms that switched to selling to the government at some point during the sample period and continued doing so. Irregular sellers comprise firms that sold periodically to the government since they switched to selling to the government in our sample.

The introduction of the latter type of comparison results in negative weights for some of the coefficients estimated.

We use the doubly robust estimator proposed by Sant'Anna and Zhao (2020) as generalized by Callaway and Sant'Anna (2021) to a setting with multiple periods and multiple treatment cohorts. The CS estimator combines the outcome regression approach and the inverse probability weighting (IPW) using pre-treatment characteristics to either condition parallel trends or weight the control group by the probability of being treated.

More formally, the outcome of interest in the CS estimator is the average treatment effect on the treated (ATT) for each cohort of treated units g at time t

$$ATT(g, t) = \mathbb{E}[Y_{i,t}(g) - Y_{i,t}(NT) \mid G_i = g, X_i] \quad (2)$$

where $ATT(g, t)$ is the average treatment effect on the treated for group g at time t ; $Y_{i,t}(g)$ is the potential outcome for unit i at time t if first treated at time g ; $Y_{i,t}(NT)$ is the potential outcome if not treated; $G_i = g$ indicates unit i belongs to cohort g (first treated at time g); and X_i represents the vector of covariates.

Callaway and Sant'Anna (2021) define the doubly-robust estimator as

$$\hat{\mathbb{E}}[ATT(g, t)] = \frac{1}{n_g} \sum_{i: G_i=g} [\omega_{i,g,t}(X_i) \times (Y_{i,t} - \hat{\mu}_t(X_i))] \quad (3)$$

with $\omega_{i,g,t}(X_i)$ being the inverse probability weights based on the set of covariates X_i ; $\hat{\mu}_t(X_i)$ is the outcome regression estimate for the control group; and n_g is the number of units in each treatment cohort g respectively.

Put in an event study-like form, the CS estimator can be re-written as

$$Y_{i,t} = \sum_{e=-3}^3 \theta_e \times D_{i,t}^e + X_i' \gamma + \alpha_i + \lambda_t + \varepsilon_{i,t} \quad (4)$$

where $D_{i,t}^e = 1$ if firm i is exactly e periods since first treatment at time t , and θ_e represents the dynamic treatment effect at event time e (in our empirical application we include 3 pre- and 3 post-treatment periods). Since every cohort-specific ATT is already computed as a change from its own pre-treatment baseline, the pooled estimate is automatically centered on zero. No further normalization is required, so we do not have to omit a reference period (e.g. $t - 1$) as in standard TWFE event-study regressions. Last, α_i and λ_t correspond to firm- and time-fixed effects respectively.

In this event-study consistent form, the term $\theta_e = \sum_g w_g \times ATT(g, g + e)$ is the weighted average of each cohort-specific ATT.²⁰ In this way, the CS estimator ensures that θ_e represents a properly weighted average

of treatment effects. The term X_i include a set of pre-treatment characteristics which we use to improve the matching between treated and not-yet treated units: the sector of the firm, the level of exports and imports registered by the firm, and the size of the economic network of a firm as proxied by the total number of buyers and suppliers.²¹ Each dot in the resulting event-study figure is the point estimate of θ_e , i.e. the weighted average, across treatment cohorts, of the change in the outcome between calendar period t and the last period before that cohort's treatment.

The Sections that follow report the main findings of the event study analysis, by outcomes of interest. To ease the interpretation of the findings, our baseline estimations always refer to the group of firms that start selling to the government at some point, and do not stop during the sample period (i.e. the *Stayers*). We adopt this approach for several reasons. The first is that, together the *Regular* suppliers, selling continuously to the government is the most frequently observed pattern in the data, although the time limitation going forward does not allow to know whether this is true for more recently treated firms. Second, this pattern is more consistent with the definition of the treatment as an absorbing state, which is common in settings similar to ours (e.g. Alfaro-Urena et al., 2022), and makes the results comparable to other papers studying the effects of winning procurement contracts (e.g. Ferraz et al., 2015). Focusing on *Stayers* also facilitates comparison of results for *Exiters*. Since exit points to a reversal of some of the post-treatment trends, excluding *Irregular* suppliers provides a clearer understanding of how the relationship with the government shapes firms' performance. The pattern of the *Irregular* suppliers is more erratic, with a majority of such firms selling more than once, and for about 3 years each time (see Appendix Table B-5). We report results using a sample that includes both *Stayers* and the *Irregular* suppliers in the baseline estimation and uses the not yet treated as a control in Appendix D.

4. Results

4.1. Sales

Fig. 2 plots the estimates of the evolution in firms total sales (left-hand panel) and sales excluding government contracts (right-hand panel) around the first year in which they start selling to government entities. We find no significant differences in the evolution of the outcomes across the treated and control groups before the treatment,²² and a sharp increase in sales relative to the control group in the year of treatment. The difference in total sales persists over the period for which we can observe treated firms.

²⁰ The subscript e is a time marker relative to the event time for each cohort (periods since treatment), and w_g are the aggregation weights that are distributed so that more numerous cohorts "weigh" more on the aggregated dynamic ATT.

²¹ We exclude the pre-treatment proxies of the economic network when we evaluate the effect of treatment on the buyers' network.

²² Numerical estimates for the Average Treatment on Treated (ATT) for all groups across all periods are reported in Appendix Section E, alongside the cohort size in each exercise proposed in this and the next sections.

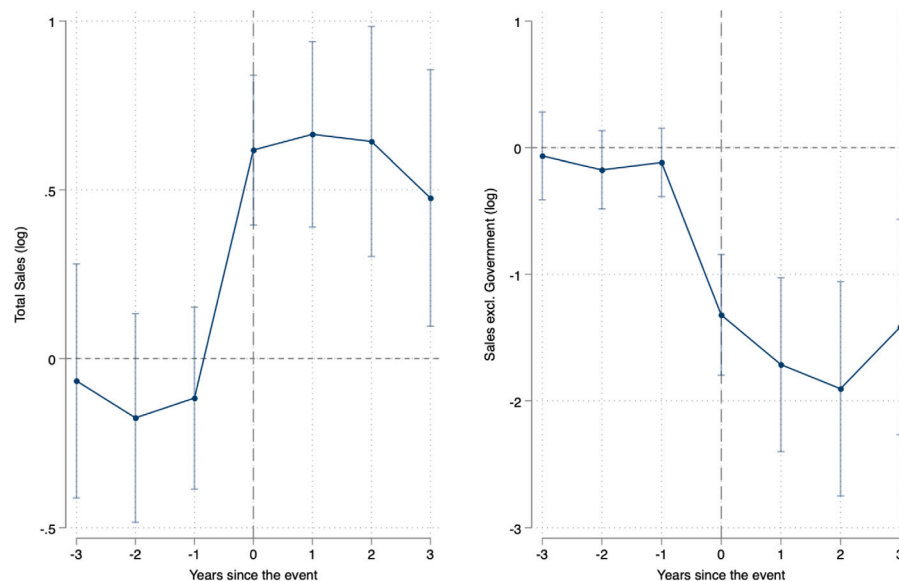


Fig. 2. Sales — *Stayers* only.

Notes: The event study compares firms that start selling to the government and continue doing so afterward to firms that will sell continuously to the government in the later years. Callaway and Sant’Anna (2021) correction is applied to account for heterogeneity in treatment timing. Sales outcomes are reported in $\log s + 0.01$.

The point estimate of 0.64 at event-time 0 implies an increase in total sales of about 90 percent ($((e^{0.64} - 1) \times 100)$) relative to the counterfactual level absent treatment. Using the mean pre-treatment sales of treated firms (UGX 3564 million, or roughly USD 0.98 million), this corresponds to an absolute increase of UGX 3195 million (about USD 0.88 million) in the first treatment year. In contrast, sales excluding government decline substantially at treatment. The point estimate of -1.26 at event time 0 implies a reduction of about 72% ($((e^{-1.26} - 1) \times 100)$), which corresponds to a loss of about 2554 million UGX (USD 0.70 million), leaving non-government sales at approximately UGX 1010 million (USD 0.28 million). The loss of sales to non-government buyers is not simply a short-term effect; sales do not begin to recover until three years after firms start selling to government entities.

We can exploit the granularity of the transaction level data to unpack sales to the private sector into (i) the total number of buyers and (ii) the average size of transactions. Results for private sector sales are plotted in Fig. 3. After the first transaction with the government, there are visible changes within the network: the number of private buyers increases (left-side panel), but the average size of transactions falls (right-side panel). Although the decline in the average size of transactions is partly mechanical, it also reflects a change in the composition of the firm’s private buyers. Additional analysis presented in Appendix Figure G-1 shows a systematic reallocation across buyer types. Specifically, firms experience a persistent decline in the number and share of pre-treatment private buyers classified as large, defined as private firms whose average purchase from a seller in a given year exceeds the seller’s median buyer–year purchase size across all years, following treatment. While firms do add new large private buyers following treatment, this extensive margin expansion is insufficient to offset the contraction among pre-existing relationships with large private buyers. As a result, the overall share of large private buyers among a firm’s customers declines following treatment. These compositional changes suggest that firms drop big private buyers to create slack to cater to the government. The findings are consistent with the survey, suggesting this pattern reflects the greater attractiveness of selling to the government (we discuss this further in Section 6).

4.2. Productivity

As in our initial descriptive analysis (Appendix Table C-1), labor productivity, measured as sales per employee, increases for firms that

sell to government entities (Fig. 4, left-hand panel). However, we no longer find a corresponding increase in value-added per employee (Fig. 4, right-hand panel).²³ This might reflect an increase in total sales and revenues without a corresponding increase in capacity or efficiency.²⁴

If productivity growth is driven by efficiency gains, we should observe a corresponding improvement in wages. When we replicate our event study-based analysis using compensation per employee as an outcome, we do not find evidence of a significant improvement. In Section 6, we explore potential channels that could underlie this finding, including capacity expansion constraints and price differences between contracts for the public sector and those with private buyers.

4.3. Losing government contracts

Fig. 5 reports the results for all the outcomes of interest in a setting in which a firm is considered treated when it *stops* selling to the government and does not obtain a contract again in the future (the “Exiters”). In this context, the treatment group includes firms in our sample period that commence selling to a government entity but that subsequently stop doing so at some point, while the control group includes firms that will stop selling to the government in future.

Results largely mirror our main findings in terms of the immediate effect, although the effect for total sales and sales to the private sector is transitory. Thus, in the year when firms exit, their total sales decline by approximately 300 million UGX, computed as $(e^{-0.5} - 1) \times 763.26$ million UGX. This temporary decline in overall sales following the loss of government demand coincides with a sharp but short-lived increase in sales to private firms. Specifically, private sales rise by roughly UGX 2420 million UGX in the treatment year, computed as $(e^{1.65} - 1) \times 574.96$ million UGX. However, this spike is transitory and does not persist

²³ Numerical estimates of the average ATT for all groups across all periods are reported in Appendix Section E.

²⁴ Note that the sample is substantially reduced compared to the one available when focusing on sales. To exclude the possibility that the lack of a significant effect on VA per employee is driven by sample selection, we replicate the event study for total sales and sales excluding the government (Fig. 2) on the sub-sample of firms for which we can match data from the VAT and PAYE. The resulting event study plot is reported in Appendix Figure D-1 and it remains consistent with our main finding.

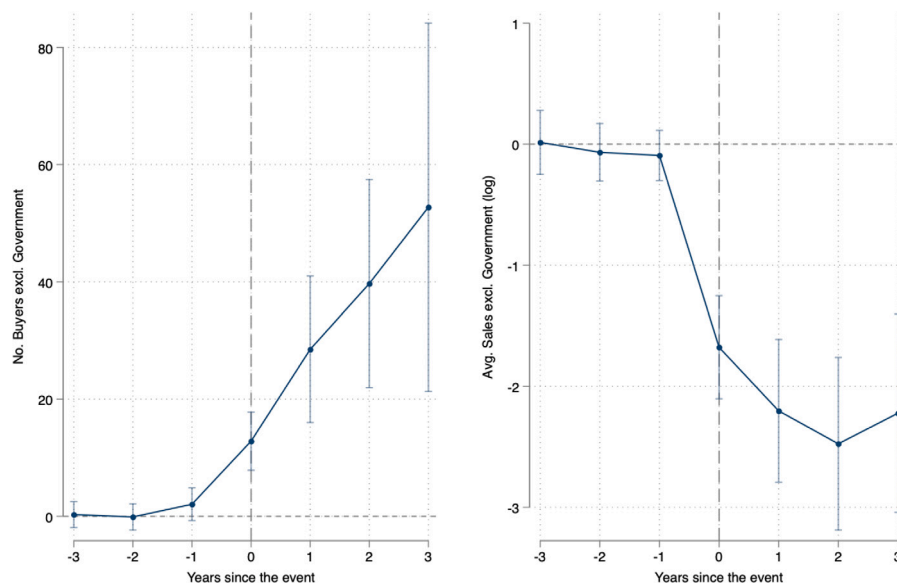


Fig. 3. Linkages.

Notes: The event study compares firms that start selling to the government and continue doing so afterward to firms that will sell continuously to the government in the later years. Callaway and Sant'Anna (2021) correction is applied to account for heterogeneity in treatment timing. The number of non-government buyers is reported in absolute values while the average size of the transaction with non-government buyers is reported in $\log s + 0.01$.

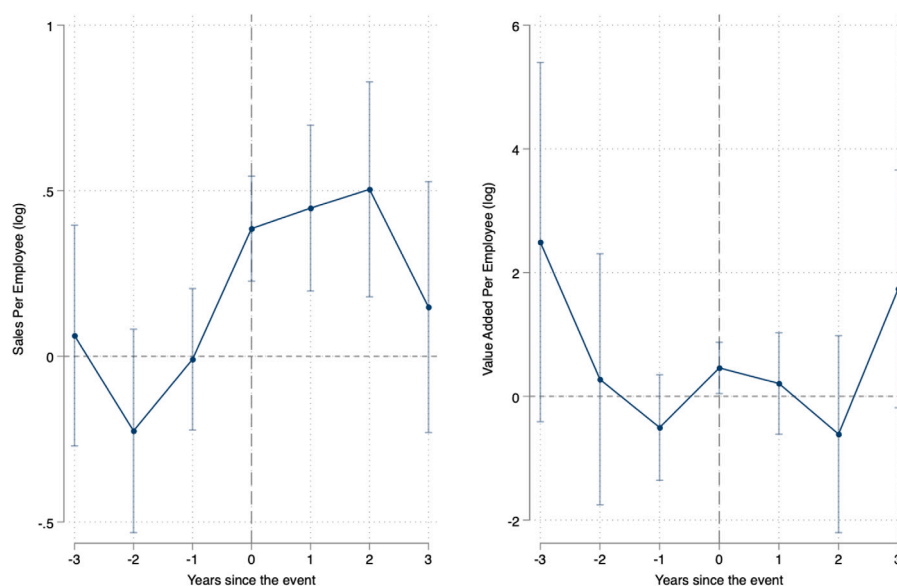


Fig. 4. Productivity.

Notes: The event study compares firms that start selling to the government and continue doing so afterward to firms that will sell continuously to the government in the later years. Callaway and Sant'Anna (2021) correction is applied to account for heterogeneity in treatment timing. Both sales and value-added per employee are reported in $\log s + 0.01$.

beyond the immediate treatment period. Firms also adjust by reducing the number of private sector interactions, but as in the case of sales, this is transitory. Firms losing government demand increase the size of transactions with private buyers, but the effect, while sizable, is short-lived. Exit also leads to a temporary decline in productivity measured by sales per employee, with no corresponding effect on value added per employee.

4.4. Selling to other large buyers

An important question in interpreting our findings is to understand whether they are specific to selling to the government or generalize to instances involving large non-government buyers. To assess this, we

replicate the analysis by adopting two alternative definitions of the treatment, based on either (i) the first transaction with a large domestic non-government buyer, defined as one whose average income over the sample period was in the top 10th percentile of the distribution (using the CIT records); or (ii) a foreign buyer based in Uganda. For consistency with previous analysis, the treatment group only includes firms that record a sale to a large (or a foreign) buyer sometime after first appearing in the sample and continue recording sales thereafter, while the control group is restricted to “not yet treated”. Results are reported in Fig. 6. The drop in sales to other private partners is unique to instances in which the government is a buyer. No such relationship is observed for firms that begin selling to large domestic private buyers or to foreign firms. This finding is not due to large domestic firms and

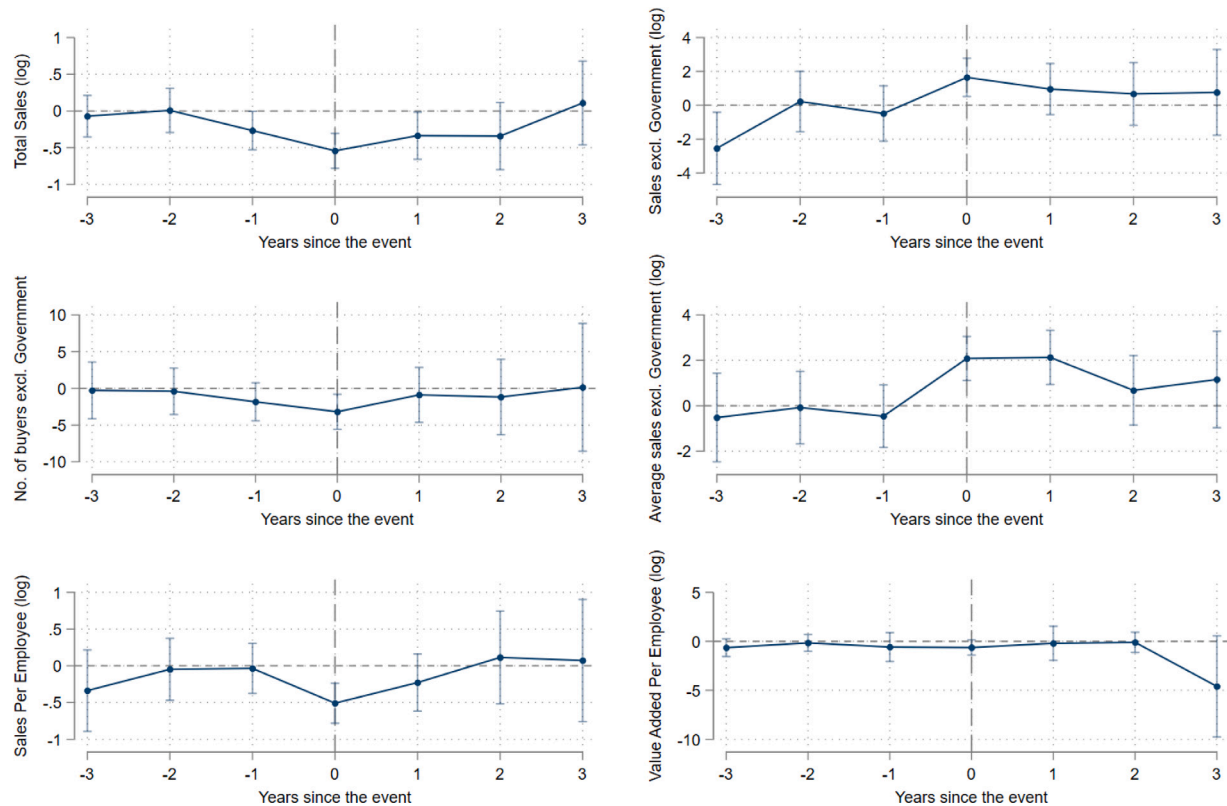


Fig. 5. Exit.

Notes: The event study compares firms that stopped selling to the government to firms that will do so in future. Callaway and Sant'Anna (2021) correction is applied to account for heterogeneity in treatment timing. Total sales, sales excl. govt., average sales to the private sector, sales per employee and value-added per employee are reported in $\log s + 0.01$. The number of non-govt. buyers is reported in absolute value.

foreign firms being less relevant buyers than the government entities. In related estimation samples, average sales to the government (9% of total sales) are not significantly greater than the average sales to large domestic firms (10% of total sales) or the average sales to foreign firms (5% of total sales).

5. Robustness checks

Alternative estimators. We check the consistency of our findings using an alternative event study estimator based on (De Chaisemartin and d'Haultfoeuille, 2020). Their estimator is a weighted average of the DiD estimands that compare the evolution of the mean outcome in two groups: those switching from no treatment to treatment between $t-1$ and t , and those remaining untreated. The estimator assumes that the treatment is an absorbing state, that is, there are no groups that switch from treatment to no treatment. A limitation of the (De Chaisemartin and d'Haultfoeuille, 2020) DiD estimator is that it does not allow matching the treated and control units based on pre-treatment characteristics. To better compare the results to our original approach, we also replicate the event study estimates by using the Callaway and Sant'Anna (2021) estimator, but without using the pre-treatment covariates for matching. The event study plots for both estimators are reported in Appendix Figures F-1 and F-2. The two plots visually confirm the main findings presented in Section 4.1, suggesting that our results do not depend on the choice of the estimator.

Sample. Fewer firms report both VAT and PAYE data than those reporting VAT only (see Appendix Table B-3). Thus, the sample used to estimate the role of Government contracts on sales and linkages (which are based on bilateral VAT data) is different from the sample used to estimate the effect on productivity (which are based on VAT and PAYE data). The difference in the sample composition might affect the results,

as some of the firms that must register for the VAT are not obliged to report on the other forms.²⁵ Hence, we replicate our event study analysis on the sub-sample of firms that report both VAT and PAYE data. The sub-sample analyses presented in Appendix Figures D-1 to D-3 are consistent with our baseline event study results.

Alternative definitions of the treatment and control groups. In Appendix D we report two sets of results in which the sample changes as follows. In Figures D-4–D-6 we consider the *Stayers* and *Regular* suppliers together, with the control group made up of not yet treated firms. Figures D-7–D-9 augment the previous exercise by adding the never treated firms to the control group. These exercises are useful in understanding whether the choices we made about the sample of the treatment and control groups influence the findings. The evidence suggests this is not the case, though with some nuances. Including the group of *Irregular* suppliers yields patterns broadly consistent with our main findings, with the exception that total sales decline in subsequent periods, a result that aligns with the patterns observed when firms stop selling to the government in Section 4.3.

Results based on a larger control group (Figures D-7–D-9) broadly reflect the direction of the estimates in our main specification. However, most event-study graphs exhibit evidence of pre-trends, which is not surprising given the substantial differences observed between treated firms (at $t-1$) and never-treated firms (see Table 2).

Misreporting. The reallocation in firms' sales that we observe in our analysis might be due to a change in their reporting behavior when they start selling to the government. For instance, firms might expect

²⁵ VAT reporting applies to most legal forms of firms, while PAYE applies to employees whose monthly pay exceeds a certain threshold.

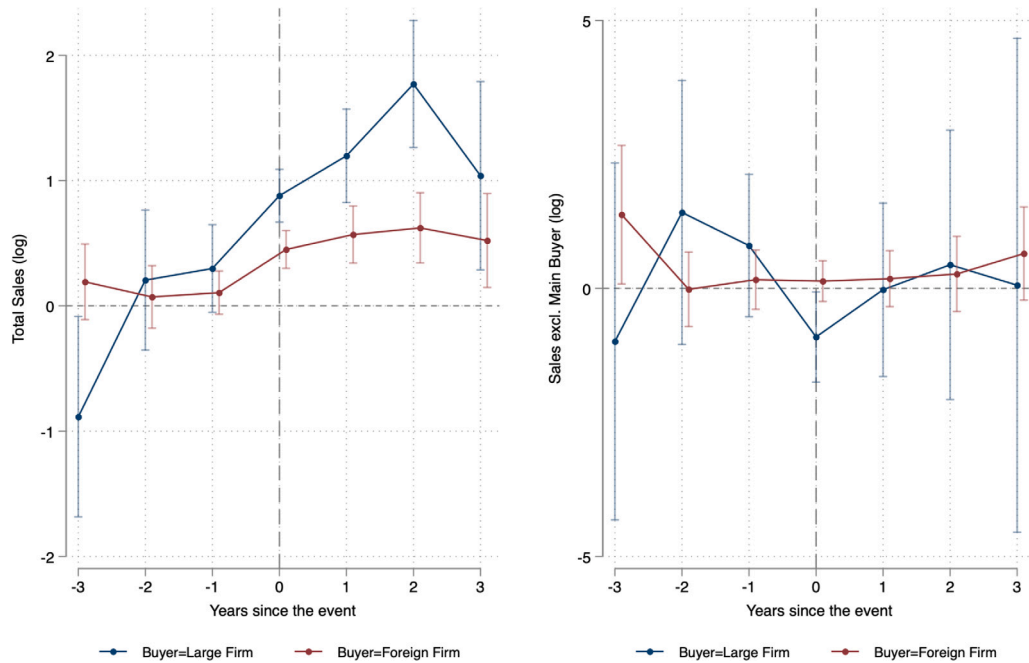


Fig. 6. Large private buyers.

Notes: The event study compares the change in outcomes due to selling to the govt. against two alternative treatments: (1) Selling to large domestic buyers and (2) Selling to foreign firms. We exclude large domestic and foreign firms when computing the number of non-govt. buyers when we estimate the effect of selling to large domestic firms or foreign firms, respectively. The (Callaway and Sant'Anna, 2021) correction is applied to account for heterogeneity in treatment timing. Both Total sales and Sales to private buyers are reported in $\log s + 0.01$. See Figure H-1 in the Appendix for the event study plot for firms average sales, number of buyers (incl. and excl. the government), and sales/value added per employee.

greater scrutiny of their VAT disclosures, and therefore begin reporting their sales more precisely. This might explain a persistent increase in overall sales when they start selling to the government, even though it cannot explain a corresponding decline in sales to the private sector. Alternatively, even if firms do not expect more scrutiny, they may want to limit their tax payments in expectation of higher revenues and therefore may decide to under report their sales to the private sector as they begin selling to the government. There is evidence that a large share of registered firms in Uganda misreport their sales and purchases to reduce tax payments (Almunia et al., 2024).

To check whether our results are affected by misreporting, we use the methodology suggested by Almunia et al. (2024). Specifically, we construct a proxy measure of misreporting, which is the difference between seller i 's reported sales (S) to domestic buyer j in year t and buyer j 's reported purchases (P) from seller i in the same year, treating non-reporting by either side as a zero. We then take the average of the discrepancies across all buyers to construct a seller misreporting measure in a given year:

$$d_{ijt} = (S_{ijt}) - (P_{jit}), \quad (5)$$

$$\bar{d}_{it} = \frac{1}{J_{it}} \sum_{j=1}^{J_{it}} d_{ijt} \quad (6)$$

Approximately three-quarters of discrepancies arise because one side does not report any transaction in a year. Notably, around one-third of these non-reporting cases are driven by buyers that never report any purchase throughout the sample period. Such buyers are plausibly below the VAT registration threshold, suggesting that a substantial share of discrepancies may reflect tax exemptions rather than misreporting behavior. Consistent with this interpretation, Mascagni et al. (2023) discuss discrepancies driven by VAT-exempt buyers in Rwanda and restrict their analysis to buyers that report a VAT declaration at least once.

We therefore present event-study estimates of discrepancies around firms' starting to sell to the government both for the full sample and

after excluding seller-buyer-year observations involving buyers that never report any purchase during the sample period (Fig. 7). Across both specifications, we find no statistically significant change in the discrepancy measure following treatment. If anything, patterns are even less clear once excluding potentially VAT-exempt buyers.

6. Mechanisms

The foregoing analysis generates four findings. First, firms that begin to sell to the government tend to grow larger. Second, there is an increase in revenue-based labor productivity, but no changes in value-added productivity. Third, selling to government buyers is associated with a reduction in sales to the private sector and a re-organization of the domestic supply chain towards more buyers but smaller transactions. Finally, firms that exit from selling to government entities do not attain previous levels of total sales, i.e., the loss of government sales is not offset by increasing sales to private companies.

These findings raise several questions. What makes the government a preferred buyer relative to private customers and what prevents firms from keeping relationships with other private buyers at scale once they start selling to the government? What can explain the absence of improvement in efficiency, proxied by value added per worker in our analysis? To gain greater insight into the mechanisms that underlie our findings we undertook an ad-hoc survey of Ugandan firms that participate in public procurement bids. The goal was to gather qualitative information on their strategies, with a focus on assessing whether and why firms shift away from sales to private buyers following transactions with the government to motivate additional analysis using available administrative data sources.

6.1. Qualitative firm survey evidence

The survey was designed and piloted in the fall of 2023 and conducted in the first quarter of 2024, targeting firms with experience with public procurement. The sampling frame was based on a list of firms

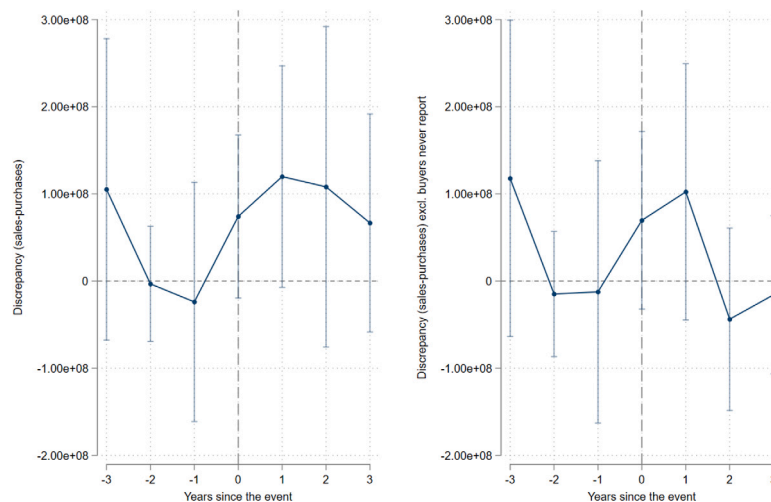


Fig. 7. Discrepancies in domestic sales.

Notes: Baseline compares early participants in procurement to late participants in procurement. Discrepancy is a proxy for the extent of misreporting in sales reported by a seller in a year. Callaway and Sant'Anna (2021) correction is applied to account for heterogeneity in treatment timing.

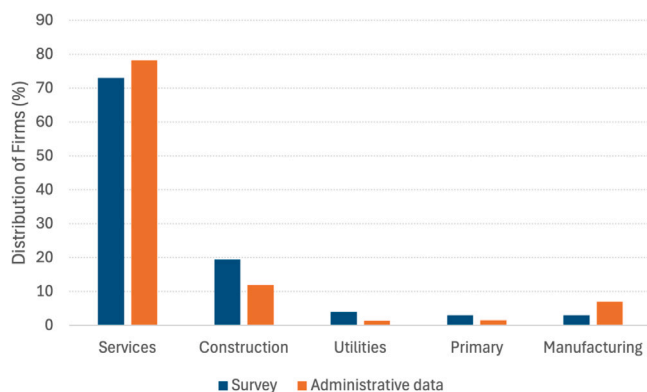


Fig. 8. Sectoral distribution of firms (surveyed firms vs. administrative dataset).

Notes: The figure compares the sectoral composition of firms in Uganda's administrative data to the sectoral composition of procurement firms in our survey.

that pre-registered with the Ugandan Public Procurement and Disposal of Public Assets Authority (PPDA). We focused on firms located in the Greater Kampala area (Kampala, Wakiso, and Mukono) that were awarded at least one government contract during the period 2017–2022 and excluded firms whose contact details provided to the PPDA were no longer valid. Additional details on the survey, including on the sample strategy and the questionnaire, are reported in Section K in Appendix.

The selection procedure resulted in a sample of 282 firms, of which a total of 236 firms were successfully interviewed. The sectoral distribution of the firms in the survey sample is similar to that in the VAT database, with most firms active in services sectors (Fig. 8). The survey sample comprises a somewhat higher share of firms in construction, while manufacturing firms are slightly underrepresented relative to the VAT data. Consistent with our main findings, 49.7% of the firms interviewed see the government as their main client, and 85.7% regard government entities as potential buyers of their goods or services. Almost 95% of respondents obtained more than one government contract over the period considered, and 54.7% indicated their firm regularly supplies government entities. Appendix Table K-1 provides summary statistics for the surveyed firms.

Government purchases on average accounted for approximately half of the respondents' turnover in the previous year (Appendix Figure K-1). Additionally, firms that define themselves as (a) continuous; or (b) frequent government suppliers tend to be larger than firms that see the government as a marginal source of demand (i.e. that only sell to the government once over the period of interest). Government entities tend to be larger buyers (91% of respondents), provide longer-term contracts (78%), and pay higher prices (64%).

6.2. Analysis of potential mechanisms

We draw on the survey findings, as well as on additional data analysis, to evaluate two potential explanations for why sales to private buyers fall following entry into government procurement: (1) capacity constraints; and (2) price differentials across public and private sector buyers.

6.2.1. Capacity constraints

Two-fifths of respondents (42%) report that their firm expanded production following award of a government contract, indicating that for many of the surveyed firms capacity constraints were not a major factor (Fig. 9).²⁶ Of the firms that did not expand, three-quarters (74%) declared they were already operating at full capacity (Fig. 9). Respondents for these firms also mentioned lack of demand and competition as other factors that restricted investment. Only 5% of firms cited access to credit as an impediment to expanding production capacity.

We cannot directly evaluate to what extent capacity constraints drive the decline in sales to private buyers that we estimate using the VAT data, but we can use the administrative data to provide supportive evidence by differentiating between firms based on sector of activity and associated average capital intensity of production. Services firms comprise a majority of those expanding capacity (62%). Firms engaging in relatively low-skill labor-intensive activities (e.g., cleaning services) will tend to have lower fixed asset ratios than manufacturing or mining activities. In principle, they may therefore be less constrained in expanding production in response to a positive demand shock. We investigate this hypothesis by exploiting the variation in capital

²⁶ Among the firms that expanded capacity following a government contract, 55% invested in their workforce (hiring more employees), while 71% invested in assets (Figure K-2 in Appendix). Shares sum up to more than 100% since some firms did both.

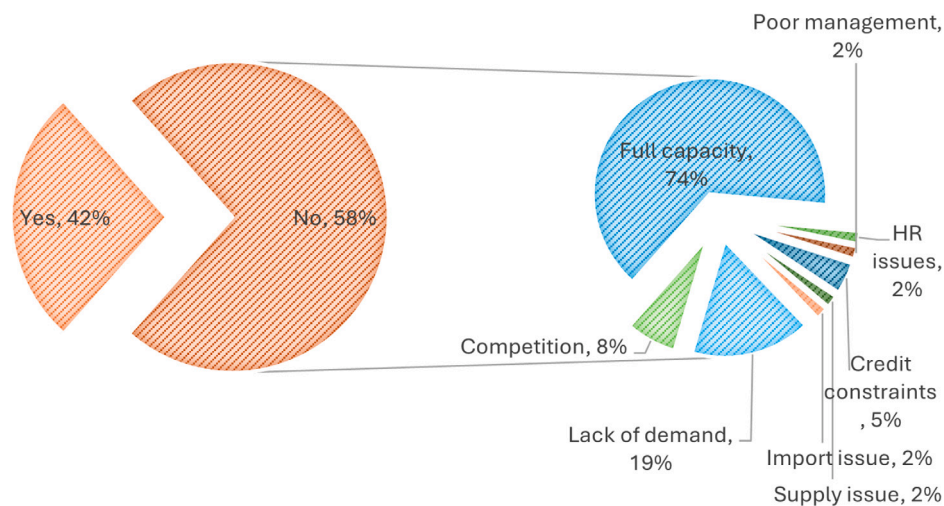


Fig. 9. Expanding capacity following government contracts.

Notes: The left side pie chart reports the responses to the following question: “Regarding your last contract awarded, did you expand production after obtaining the contract?”. The right side pie chart lists reasons given for not expanding production (multiple choices were possible). Only firms responding “no” to the first question were asked to respond to the second question. Overall, 217 firms responded to the first question and 125 to the second.

intensity across sectors under the assumption that firms in sectors with higher fixed asset ratios may be more constrained in expanding output in the short run if capacity utilization rates are high. Firms in services are less capital intensive (see Appendix Figure J-1), making it easier to expand in the short run after obtaining a contract to supply government entities. Fig. 10 (left hand panel) shows that total sales increase by similar magnitudes for both services and non-services firms. However, the adjustment margin differs. Services firms, which face less constraints on expanding capacity, increase total sales without reducing private sector sales (right hand panel). In contrast, non-services firms, which have higher fixed assets and confront greater capacity constraints, expand sales to the government primarily by reallocating away from private sector sales. Despite these different adjustment margins, the overall growth in total sales is similar across the two groups. This suggests that non-services firms are able to achieve comparable revenue gains without expanding production to the same extent, which is consistent with government contracts offering higher prices relative to private sector sales. We explore this as a potential explanation in the next sub-section.

We next assess whether firms expand production capacity when they start selling to the government. Fig. 11 shows that selling to the government is not related to a clear expansion in total investment or fixed assets when they start selling to the government. While fixed assets exhibit an increase, the estimates are imprecise and do not reflect a discrete investment response at the time firms begin selling to a government entity. Consistent with this, the capital-labor ratio exhibits substantial volatility but no sustained post-treatment increase, and the ratio of fixed assets to sales declines following treatment.

6.2.2. Government as a more profitable buyer

As noted previously, 64% of the survey respondents state that government entities pay higher prices for their goods or services than private buyers. Related, when asked why they reduce sales to other firms following a government contract, a majority (60%) indicate this is because private firms pay lower prices. While it is not possible to directly test this channel using the VAT data due to the absence of information on prices and quantities, we provide suggestive evidence for this mechanism using three diverse strategies. First, suggestive support for the price mechanism is provided by changes in the pattern of purchases of firms selling to the government from their suppliers. If the overall sales growth observed in the VAT data is due to the government offering higher prices, the firm’s purchase of inputs

should be relatively stable. Appendix Figure I-1 shows that there is no effect on total purchases from suppliers in the year a firm begins selling to the government, and while we observe an increase in later years, these effects are imprecisely estimated. Further, there is no significant increase in the number of suppliers from the time a firm starts selling to the government and continues doing so. Second, we exploit information from Customs clearance transactions data. These data include information on the quantity and the value of goods that are exported and imported by a given firm. We focus on the group of firms in the “Wholesaler, Retailer and Repair” sector where input quantities, proxied by imports, can be expected to be highly correlated with sales, as their activity does not generally involve transformation. If a firm imports most of what it sells, the quantities imported should approximate the quantities that are sold, and any difference in the value of sales after beginning to sell to the government should then reflect a change in prices. We use two thresholds for the share of imported inputs in total purchases: (i) greater than or equal to 65%, and (ii) greater than or equal to 95%. Table 3 reports results of an OLS regression estimating the relationship between the value of sales and import quantities subsequent to a firm in the “Wholesale and Retail” sector obtaining a government contract. Columns (1) and (2) use the 65% threshold and columns (3) and (4) show results when using the 95% threshold ratio of import value to total sales. There is a positive relationship between commencing to sell to the government and the value of total sales, but not for imported quantities. This suggests that sales to government entities are associated with higher prices. Third, we exploit profitability data from firms’ CIT records and apply the baseline event-study framework to examine how reported profits evolve when firms begin supplying the government. Fig. 12 shows an increase in profit levels once firms start selling to the government. This additional evidence is again consistent with the interpretation that selling to the government is more profitable.

6.2.3. Price differentials, capacity constraints and uncertainty

While higher prices for government contracts appear to play a role in the observed reallocation of supply capacity away from private buyers, there is also evidence that capacity constraints may be pertinent. A potential explanatory factor for both decisions not to expand capacity when this involves fixed costs (investment in tangible assets) and for the observed price differentials is uncertainty associated with the procurement process. Four fifths of surveyed firms (81%) report that the government is unlikely to pay them on a timely basis. When

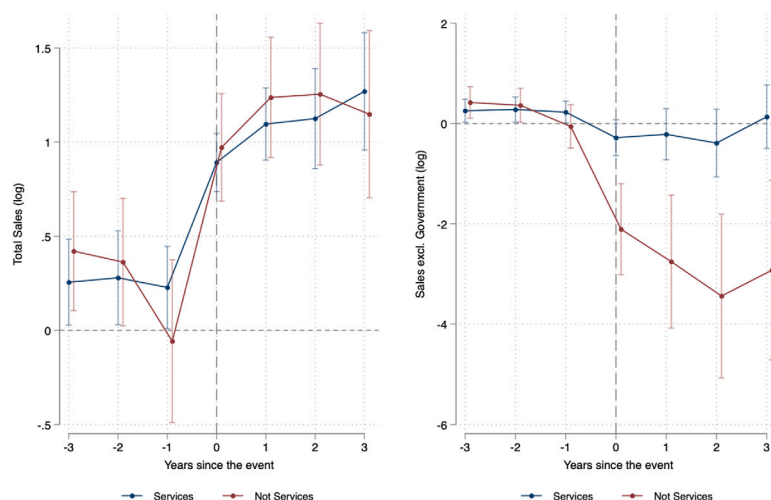


Fig. 10. Services vs. non-services firms: Total Sales and Sales excluding the Government.

Notes: Baseline compares early participants in procurement to late participants in procurement. The change in total sales and sales excluding government is compared across Services and Non-Services sectors. Callaway and Sant'Anna (2021) correction is applied to account for heterogeneity in treatment timing. Total sales and sales excluding government is reported in $\log s + 0.01$.

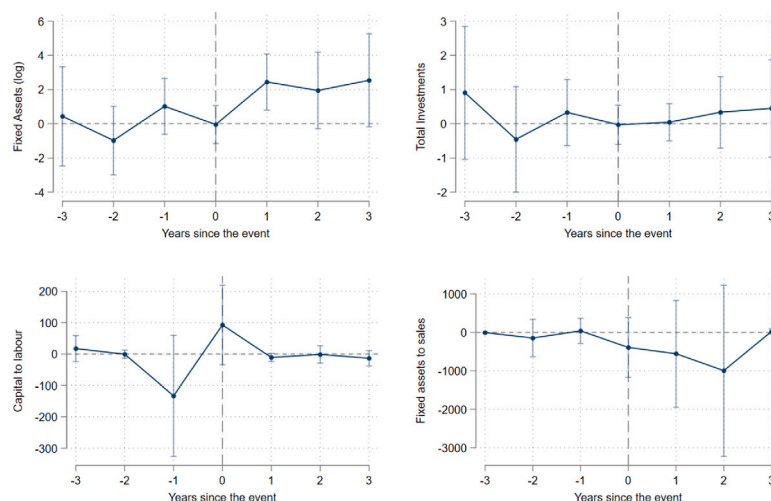


Fig. 11. Assets and investments.

Notes: Baseline compares early participants in procurement to late participants in procurement. Callaway and Sant'Anna (2021) correction is applied to account for heterogeneity in treatment timing. Outcome variables are respectively (i) fixed assets (ii) total investment (iii) capital to labor ratio and (iv) fixed assets to sales. Fixed assets and total investments are in logs.

Table 3

Value of sales to government and imported quantities (firms in the Distribution sector).

	(1) Total sales (value)	(2) Import quantity	(3) Total sales (value)	(4) Import quantity
After first sale	0.714*** (0.147)	-0.120 (0.190)	0.458** (0.230)	-0.175 (0.322)
Sample	3525	3525	1480	1480

Notes: This table reports the relationship between a dummy indicating the year in which firms in the wholesale sector (ISIC rev. 4 Section G, Divisions 45, 46, and 47) start selling to the government, their value of sales and imported quantities. Columns 1 and 2 (3 and 4) report estimates for wholesalers with a share imports in total purchases above 65% (95%). Standard errors are clustered at the firm level. Estimates include firm and sector-year fixed effects. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

asked about their *last* contract with the government, 64% of the firms stated that it was not paid on time. The average delay in payment

was five months, with a maximum delay of more than two years (Appendix Figure K-3). Such delays have potential consequences for investment planning and may have an effect on the willingness as well as the ability of firms to expand capacity. Expected payment delays also may be reflected in contract prices, insofar as firms factor this into their bids. Uncertainty regarding the duration of relationships with government entities and the prospects of renewal and obtaining government contracts over a long period of time can also be expected to influence firm decisions. This is an issue that calls for further research, as our data do not permit assessment of the potential role of uncertainty as a factor explaining investment or pricing differentials.

7. Conclusion

This paper contributes to the literature analyzing the relationship between government demand and firm performance, using detailed administrative data for Ugandan firms. Applying an event study approach that accounts for self-selection and the heterogeneity in the timing of

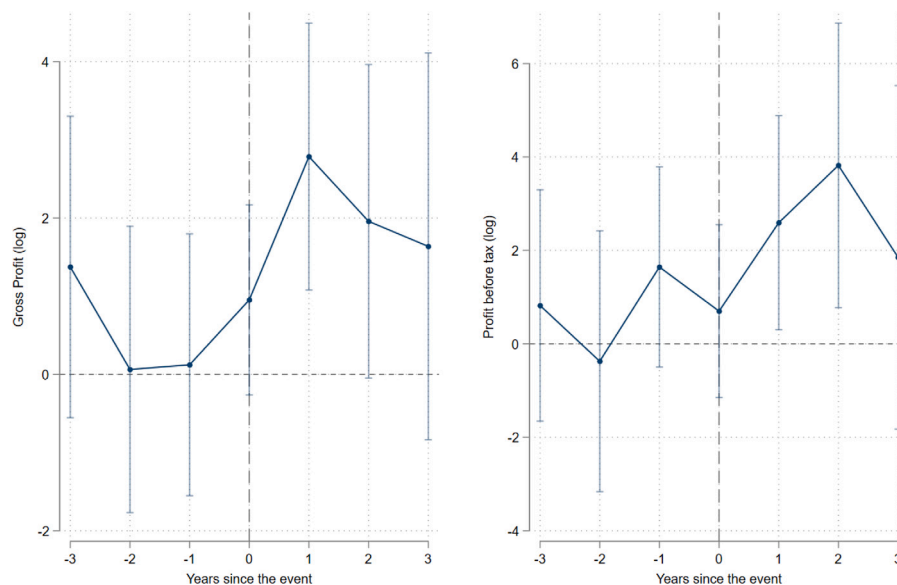


Fig. 12. Profitability.

Notes: Baseline compares early participants in procurement to late participants in procurement. Callaway and Sant'Anna (2021) correction is applied to account for heterogeneity in treatment timing. The variables “gross profits” and “profit before tax” are from the Corporate Income Tax return form. Both the outcome variables are reported in $\log s + 0.01$.

the treatment we find evidence of positive effects on sales and output per worker in firms that start to sell to government entities. The positive association with total sales is accompanied by a reduction in sales to non-government entities, i.e., overall sales growth is partly at the expense of a reallocation of firm-level supply to government buyers.

Our findings suggest that many firms are limited in their ability to expand capacity and/or that government contracts are more profitable. We find some suggestive evidence for the capacity constraint hypothesis, reflected in the reallocation of sales away from private buyers being associated with differences in factor intensity of production: it is less pronounced for firms in services that are less capital intensive and thus may more easily be able to expand output in the short run by increasing employment. While the firm-level survey provides support for the existence of capacity constraints, some 40% of respondents active in the procurement market report an increase in production capacity during the period considered. This suggests capacity constraints are not a major driver of the sustained reduction in sales to the private sector. This assessment is bolstered by most interviewees indicating that access to finance was not a major concern. Instead, most respondents reported that relative to private buyers, the government pays higher prices and that this was the main motivation for reducing sales to other firms.

Understanding the reasons underlying these patterns is important both for the design of procurement processes, an issue our paper does not engage with, and for the evaluation of the effectiveness of efforts to use public procurement as an industrial policy instrument in low-income country contexts where government demand is significant. This will require researchers to have access to both VAT bilateral transactions data and transaction-level public procurement information (contract awards).

CRediT authorship contribution statement

Bernard Hoekman: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Marco Sanfilippo:** Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Filippo Santi:** Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Rohit Ticku:** Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jdeveco.2026.103832>.

Data availability

The authors do not have permission to share data.

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When Digital Nudges Nag: Evidence on Program Take-up from a WhatsApp Intervention*

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I INTRODUCTION

Individuals from vulnerable populations often display low enrollment in public services that could substantially enhance their welfare (Currie 2006). Access to public programs is rarely automatic, as applicants must navigate stringent eligibility criteria and administrative requirements that impose disproportionate informational and compliance costs on those least equipped to bear them. Social media platforms such as WhatsApp offer a potentially cost-effective channel to reduce these frictions by disseminating timely, targeted information to large numbers of people at low cost, including those who lack access to in-person outreach or harbor distrust toward government institutions. Yet whether information nudges delivered through these channels actually raise take-up remains an open empirical question, and the conditions under which they may backfire are poorly understood. We study how informational videos distributed via WhatsApp affect take-up of a regularization program among Venezuelan migrants in Colombia, a population that is both highly mobile and difficult to reach through conventional outreach channels.

This study focuses on Colombia’s Estatuto Temporal de Protección para Migrantes Venezolanos (ETPV), a 10-year regularization program for undocumented Venezuelan migrants (herein “migrants”). Under the ETPV, migrants who arrived in Colombia before January 2021 can obtain a temporary protection permit (PPT), an identification document that provides access to healthcare, financial services, and formal employment. Previous studies of an earlier version of this program documented large improvements in income and consumption among beneficiaries, but also persistently low take-up rates (Ibáñez, Moya, Ortega, Rozo and Urbina 2025). A qualitative study conducted by our research team prior to the registration period identified three recurring barriers to enrollment: limited program awareness (Finkelstein and Notowidigdo 2019; Bhargava and Manoli 2015), distrust of government institutions (Watson 2014), and psychological aversion to procedural complexity and administrative hassle (Bhargava and Manoli 2015; Kleven and

[Kopczuk 2011](#)).¹ These findings motivated the design of three informational videos, each targeting one of the identified barriers.

We employed a randomized controlled trial to investigate the impact of three video treatments delivered via WhatsApp on PPT take-up rates among 1,375 undocumented Venezuelan migrants recruited in person on the Caribbean Coast of Colombia. Participants were randomly assigned to one of three treatment arms or a control group. Drawing on the EAST framework ([BIT 2014](#)), we designed each video to address one of the three identified barriers: an *awareness* video describing program benefits, a *trust* video narrated by a migrant who had already registered, and a *step-by-step* video providing detailed guidance on registration procedures. Treated participants received their designated video via WhatsApp on four occasions at one-week intervals. Our primary outcome is whether participants requested or attended the biometric data appointment, the final administrative step before PPT issuance.²

Surprisingly, we find negative effects of the video intervention on all three outcomes. Receiving a video reduced the intention to register by 12.2 percentage points, the probability of initiating the registration process by 7.7 percentage points, and the likelihood of requesting the PPT by 8 percentage points, representing a reduction of roughly 15 percent relative to the control group mean. Decomposing the ITT following [Abadie \(2002\)](#), we find that both effects are negative for both compliers and defiers. Compliers are individuals who watched the video when assigned to treatment and would not have watched it otherwise; among them, engagement with the informational content discouraged registration. Defiers are individuals who did not watch the video despite being assigned to treatment; among them, assignment itself generated a negative response regardless of engagement with the content.

¹The qualitative study consisted of 42 in-depth interviews with Venezuelan migrants residing in Colombian cities with a high density of migrants.

²Due to IRB restrictions and the prevalence of homonyms, administrative record matching was not feasible; outcome measures rely on self-reported registration status.

To examine who is most affected by the intervention, we implement an endogenous stratification approach following [Chernozhukov et al. \(2018\)](#), which characterizes how treatment effects vary across individuals with different baseline probabilities of registering, while avoiding the endogeneity concerns that arise when conditioning on post-treatment variables such as video engagement. The results show that negative treatment effects are not uniform across the population and are largest among individuals with intermediate predicted baseline likelihoods of registering voluntarily. This is inconsistent with a digital exclusion story, in which effects would be concentrated among the least likely to register. Instead, it suggests that the intervention was most disruptive for individuals at the margin of the participation decision, consistent with a nagging effect among those whose registration decision was most sensitive to external contact.

We consider two channels through which informational nudges reduced program take-up. The first is information backfire among individuals who engaged with the video content. Separate ITT estimates by video type show that more detailed content generates larger negative effects, consistent with the interpretation that procedural information increased perceived complexity and hassle costs. This interpretation is supported by the LATE estimates, which show that among individuals whose engagement was affected by treatment assignment, watching the video reduced take-up.

The second channel is contact frictions among non-engagers. The ITT decomposition implies an average effect of 4.3 percentage points among the 28.2 percent of treated individuals who did not watch the video. Follow-up qualitative interviews with 30 treated participants provide suggestive evidence on the nature of these frictions. Non-engagers reported frustration with repeated contact attempts, difficulties accessing video content, and technological barriers including limited internet access and unfamiliarity with digital tools. While this qualitative evidence is not causal, it is consistent with a contact friction mechanism operating independently of video content.

Overall, our results show that nudges can generate meaningful negative effects even when carefully designed with behavioral principles in mind, because individuals can react negatively to both the content and the mode of delivery. The costs of unsolicited contact can outweigh the benefits of information, particularly for those at the margin of the participation decision.

Our work contributes to the large body of work on public program take-up and its determinants. Prior research has identified information asymmetries, high learning costs, and administrative complexity as the principal barriers to enrollment ([Currie 2006](#), [Kleven and Kopczuk 2011](#)). A growing literature examines whether informational interventions can reduce these frictions, with mixed results. While some studies find that providing clear program information increases take-up ([Daponte, Sanders and Taylor 1999](#), [Manoli and Turner 2014](#), [Bhargava and Manoli 2015](#)), others document limited or negligible effects, particularly in settings characterized by complexity or low attention ([Bettinger et al. 2012](#), [Hainmueller et al. 2018](#)). We extend this literature by showing that informational interventions can generate meaningful negative effects even when carefully designed with behavioral principles in mind. Our results indicate that the sign of the treatment effect depends critically on baseline motivation and engagement, dimensions that prior work has not examined jointly.

We also contribute to the literature on behavioral nudges and their unintended consequences. The nudging literature has largely focused on whether nudges are effective or neutral, with less attention paid to the conditions under which they backfire ([DellaVigna and Linos 2022](#)). We provide evidence of a nagging effect in which repeated unsolicited contact reduces take-up among individuals already inclined to act voluntarily. This mechanism is distinct from inattention or information avoidance and operates through two channels: increased perceived complexity among engagers and contact frictions among non-engagers. Our findings suggest that the line between a helpful nudge and a counter-

bian government, and administrative bottlenecks. These patterns directly motivated the intervention we study in this paper.

II.B The ETPV Program

Before PEP expired in 2020, the Colombian government launched the Estatuto Temporal de Protección para Migrantes Venezolanos (ETPV), a ten-year regularization program for Venezuelan migrants who arrived in Colombia prior to January 2021 (see Figure A.1 for a timeline). The application process required migrants to register in an online census (the Registro Único de Migrantes Venezolanos, RUMV), upload supporting documents, and schedule an in-person biometric appointment, after which a temporary protection permit (PPT) was issued. The registration window ran from May 2021 to June 2022, and the complete process is described in Figure A.2. The PPT grants holders legal status, a work permit, access to public health services, education, the financial sector, and the ability to validate Venezuelan educational credentials. By October 2022, nearly 2.5 million Venezuelan migrants had completed the RUMV census, representing one of the largest regularization efforts in Latin American history. Figure A.3 illustrates the geographic distribution of applicants across the country.

III THE INTERVENTION

III.A Rationale

As described in Section II, our qualitative study identified three barriers to PPT registration: limited awareness, distrust, and procedural confusion.⁴ Our WhatsApp intervention targeted these barriers by providing information through a cost-effective and scalable platform.

⁴The study included 42 in-depth interviews with Venezuelan migrants residing in Colombia since 2018, spanning PEP-RAMV beneficiaries, eligible non-registrants, and individuals who participated in RAMV but could not access PEP. The main findings were that desire for official documentation was a primary migration motivation, but registration was hampered by limited program awareness, fear that RAMV would be used to deport undocumented migrants, and practical bottlenecks including weak social networks, transportation costs, and lack of incentives to register.

III.B Design

We randomly assigned a total of 1,375 eligible individuals into four groups of equal size that consisted of three treatment arms and one control group.⁵ The research team interviewed individuals selected for the intervention in person at registration to collect basic sociodemographic characteristics. The three treatment arms involved the dissemination of a different video through WhatsApp, each addressing a specific barrier to registration for the PPT such as lack of program awareness, distrust of the government, and details regarding registration that were intended to reduce procedural difficulties.

We based the intervention design on the EAST methodology developed by the Behavioral Insights Team (BIT 2014), which emphasizes making information easy to understand, attractive, social, and timely. The videos simplified information about benefits, eligibility, and the registration process, incorporated social norms messaging about community registration rates, and were strategically timed based on insights from previous research with this population.⁶

Video 1 featured a Colombian actor who portrayed an official and provided clear and concise information on program eligibility, costs, and the registration process (awareness video). Video 2, narrated by a Venezuelan forced migrant and mother of two children who had registered in the program, provided the same information as Video 1 but added anecdotal evidence about her experience with the program in order to build trust and empathy (trust video). Finally, Video 3, which had the same Venezuelan narrator as Video 2, offered more detail on the registration process with a step-by-step guide on how to register online (step-by-step video). All the video scripts are provided in the Appendix B and the design of the intervention is illustrated in Figure 1.

⁵Our initial plan was to recruit 4,180 eligible Venezuelan migrants, but we could only identify 1,375 undocumented individuals willing to participate. We revised our pre-analysis plan accordingly.

⁶The value of the incentive was 10,000 Colombian pesos (COP).

III.C Recruitment and eligibility

In partnership with Innovations for Poverty (IPA) Colombia, we recruited participants in Magdalena and Atlántico as they are home to many highly vulnerable migrants, according to the Colombian population census of 2018. The sample was collected to be representative of both Santa Marta, one of the largest urban centers, and rural areas that included Ciénaga, Sabana Larga, Fundación and Baranóa (Figure 1).⁷ We partnered with local community leaders and Venezuelan enumerators to build trust, advertise the study in areas frequented by migrants, and establish registration points in marginalized communities.

The study's eligibility criteria were carefully defined to ensure that individuals were both eligible for the program and vulnerable. Specifically, we recruited Venezuelan migrants who were of legal age (18 years or older), resident in any of the selected municipalities, undocumented, and who arrived in Colombia before January 1, 2021. Additionally, participants had to have access to a phone with WhatsApp and internet. The final sample numbered 1,375 individuals who met these criteria and agreed to participate in the study.

Table 1 presents descriptive statistics. The sample is predominantly young, female, and low-income, with an average monthly income representing 20 percent of the Colombian minimum wage. Most participants reported regular internet access, suggesting mobile-based interventions were feasible for this population.

We compared our sample with Venezuelan migrants surveyed in the 2021 Colombian Labor Force Surveys and the 2020 Venezuelan Refugee Life Study, a nationally representative sample of Venezuelan migrants in Colombia (Table A.1). Participants in our study are among the most vulnerable according to measures of income, education, unemployment, and access to health services, consistent with our focus on regions with low regularization rates.

⁷We did not recruit individuals in Barranquilla due to costs.

III.D CONCEPTUAL FRAMEWORK

This subsection provides a framework for interpreting how an informational intervention can reduce program take-up, depending on whether individuals engage with the content or not.

Setup. Let $Z_i \in \{0, 1\}$ denote assignment to treatment (receiving informational videos via WhatsApp), $T_i \in \{0, 1\}$ denote engagement with the informational content (e.g., watching the video and obtaining accurate program information), and $Y_i \in \{0, 1\}$ denote take-up (e.g., initiating or completing registration). Our baseline estimates capture the intention-to-treat (ITT) effect:

$$ITT = \mathbb{E}[Y_i \mid Z_i = 1] - \mathbb{E}[Y_i \mid Z_i = 0].$$

Heterogeneous engagement and behavioral responses. We allow treatment assignment to affect both engagement and take-up, but we do not assume that assignment always increases engagement. Some individuals may actually be less likely to engage with informational content when assigned to treatment than they would have been otherwise, for instance because unsolicited contact generates avoidance or disengagement. Individuals who respond to treatment in the opposite direction from its intent are called defiers, and their presence is central to understanding why the intervention reduced take-up on average.

Two channels. We consider two channels through which the intervention may reduce take-up. The first operates among individuals who engage with the informational content. For these individuals, exposure to the videos may update beliefs about the complexity, time requirements, or administrative burden of the registration process. When information increases perceived costs in this way, the intervention reduces take-up among those who engaged with it most directly. The second channel operates among individ-

uals who receive but do not engage with the content. For this group, the mere receipt of unsolicited messages may generate friction, induce annoyance, reduce trust, or trigger avoidance, lowering take-up even without any information acquisition.

Link to the Intent-To-Treat (ITT) Estimates. Individuals in the sample differ in how their engagement responds to treatment assignment. Compliers engage with the content when assigned to treatment but would not have otherwise. Defiers do the opposite, disengaging precisely because they were contacted. When both types are present, the overall ITT reflects a combination of their responses, and a negative average effect can arise even when both compliers and defiers experience negative effects. To characterize these patterns, we complement the analysis with bounds on the share and effects of defier-type responses following [Kowalski \(2023\)](#) and an endogenous stratification approach that allows treatment effects to vary with baseline take-up propensity [Abadie \(2002\)](#).

Testable predictions. The framework yields two core predictions. If informational content increases perceived costs, treatment effects should be negative among engaged individuals (i.e., compliers). If unsolicited contact generates frictions, the negative ITT should also be driven by individuals who receive but do not engage with the content (i.e., defiers).

IV DATA

After in-person recruitment (which included a short sociodemographic baseline survey) and randomization, participants were contacted via WhatsApp up to four times to deliver the intervention and collect outcome data. Individuals assigned to treatment received a video link followed by a short survey approximately one hour later, while the control group received only the survey. Participants were recontacted only if they had not responded in previous attempts and had not yet requested the biometric data appointment, which we use as the primary proxy for program take-up.

We define take-up as requesting or attending the biometric appointment. Attendance

V.A Intention-to-Treat effects

We estimate the effects of the intervention on three outcomes: (i) intention to register for the PPT, (ii) initiation of the registration process (defined as starting the RUMV registration), and (iii) program take-up (defined as requesting or attending the biometric data appointment, or receiving the PPT). Outcome information corresponds to the last WhatsApp contact with the participant.

Our baseline estimates capture the intention-to-treat (ITT) effect of assignment to receive informational videos (Z_i) on take-up outcomes (Y_i). These estimates reflect the effect of being assigned to treatment regardless of whether individuals engage with the content (T_i). We estimate the following specification:

$$Y_i = \alpha + \beta \text{Any Video}_i + \varepsilon_i \quad (1)$$

where Y_i denotes the outcome of interest for individual i , and Any Video_i is an indicator equal to one if the individual was assigned to receive any informational video (i.e., Z_i). Standard errors are robust to heteroskedasticity. We also estimate specifications that disaggregate effects by video type to assess whether responses differ across the awareness, trust, and step-by-step videos.

V.B Heterogeneous Effects

To examine treatment effect heterogeneity, we implement an endogenous stratification approach following [Abadie \(2002\)](#) and [Chernozhukov et al. \(2018\)](#). This approach characterizes how treatment effects vary across individuals with different baseline likelihoods of registering, while avoiding the endogeneity concerns that arise when conditioning on post-treatment variables such as video engagement.

We proceed in three steps. First, we use the control group to estimate a model predicting take-up as a function of baseline covariates, implementing leave-one-out cross-validation

tempt. Panel A reports the first-stage relationship between assignment to receive a video and engagement with the informational content. Assignment increases the probability of watching the video by 71.8 percentage points, indicating that while a large majority of treated individuals comply with assignment, roughly 30 percent do not engage with the video content.

Panel B reports intention-to-treat estimates. Assignment to receive any video reduces take-up across all three outcomes. The probability of requesting the PPT decreases by 8 percentage points relative to a control mean of 54 percent, a 15 percent decline. Starting the registration process decreases by 7.7 percentage points relative to a control mean of 59 percent, a 13 percent decline. Intention to register decreases by 12.2 percentage points relative to a control mean of 83 percent, a 15 percent decline. These are sizable reductions, indicating that the intervention meaningfully discouraged participation at every stage of the registration process.

Panels C and D report instrumental variables estimates (i.e., LATE, Local Average Treatment Effects) that isolate the effect of actually watching the video, as opposed to merely being assigned to receive it. Panel C uses a single instrument (assignment to any video) and Panel D uses the three video-type assignments as separate instruments. The LATE estimates are stable across both specifications. Among individuals whose watching behavior was changed by assignment, viewing the video reduces the probability of requesting the PPT by 9.5 percentage points (Panel C) and 8.7 percentage points (Panel D), representing a 16 to 18 percent decline relative to the control mean. Effects on registration initiation are similar in magnitude, while effects on intention to register are larger, ranging from 13.6 to 14.4 percentage points. That these estimates are negative indicates that the backfire effect is not driven entirely by non-watchers reacting to unsolicited contact. The video itself discouraged take-up among those who watched it.

Panel E decomposes the ITT into the contributions of two groups, following [Abadie](#)

VIII EXPLORING POTENTIAL CHANNELS FOR THE TAKE-UP REDUCTIONS

This section examines further why the intervention reduces take-up. Guided by the conceptual framework, we consider two channels: (i) information backfire among individuals who engage with the informational content, and (ii) contact frictions among individuals who do not engage.

VIII.A Information Backfire Among Engagers

More detailed content generates larger negative effects. Appendix Table [D.2](#) disaggregates the reduced-form effects by video type. All three videos reduce take-up, but the magnitudes vary systematically with content complexity. The step-by-step video, which provides the most detailed procedural guidance, generates the largest effects across all outcomes, reducing PPT requests by 11.7 percentage points, registration initiation by 10.8 percentage points, and intention to register by 16.8 percentage points. The awareness and trust videos produce smaller and in some cases insignificant effects on behavioral outcomes, though both significantly reduce intention to register. This result across video types is consistent with the interpretation that more detailed procedural information increases perceived complexity or hassle costs, amplifying the backfire effect among those who engage with the content.

These findings are consistent with the LATE estimates reported in Table [3](#), which show that, among individuals whose engagement is affected by treatment assignment, watching the video reduces take-up. Together, the evidence suggests that exposure to detailed information may increase perceived complexity or hassle costs, leading individuals to opt out of the registration process.

VIII.B Contact Frictions Among Non-Engagers

Non-engagers drive large negative effects. Appendix Table [E.1](#) separates treatment effects by whether the individual watched the video. Among treated individuals who did not watch ($N = 427$), assignment reduces the probability of requesting the PPT by 25.7 per-

centage points, starting registration by 26.5 percentage points, and intention to register by 34.2 percentage points, all relative to the same control means. Among those who watched ($N = 1,002$), effects on behavioral outcomes are small and statistically insignificant, with a reduction of 4.8 percentage points for PPT request and 4.3 percentage points for registration initiation. These patterns are descriptive rather than causal, as individuals who chose not to watch the video when assigned to treatment are not a random subset of the treated sample, and the comparison between watchers and non-watchers reflects selection into engagement as much as the effect of content exposure. Nonetheless, the concentration of negative effects among non-engagers is consistent with the contact friction channel, in which unsolicited outreach generates avoidance or disengagement before any informational content is encountered. The decomposition in Panel E of Table 3 provides a cleaner quantitative bound on this channel, implying an average effect of -4.3 percentage points among the 28.2 percent of treated individuals who did not engage. Overall, the evidence points to contact frictions among non-engagers as a quantitatively important driver of the overall take-up reduction.

What did non-engagers say? We carry out approximately 30 qualitative interviews with participants who were assigned to treatment to inquire about the reasons behind the negative effects. The results suggest that non-engagers report frustration associated with repeated contact attempts, difficulties accessing video content, and technological barriers such as limited internet access or lack of familiarity with digital tools. Some participants also report that receiving messages without engaging with the content leads to confusion or disengagement.

Overall, the evidence supports two channels through which the intervention reduces take-up. Among engagers, exposure to detailed procedural information reduces take-up. Among non-engagers, assignment itself introduces frictions that discourage participation.

IX DISCUSSION

Providing informational videos via WhatsApp to undocumented Venezuelan migrants reduced program take-up by 8 percentage points, a 15 percent decline relative to the control mean. The negative effect operates through two channels. Among individuals who watched the video, exposure to detailed procedural content increased perceived complexity and discouraged registration. Among those who received but did not engage, unsolicited contact generated friction that similarly reduced take-up. The concentration of effects among individuals at intermediate predicted take-up levels, together with the immediate emergence of negative responses at first contact, points to information backfire rather than contact fatigue as the primary driver.

These findings carry direct implications for the design of digital outreach programs targeting vulnerable populations. Expanding informational content does not reliably increase participation and may actively reduce it when content raises perceived complexity; programs should prioritize simplifying over expanding informational scope. Those at the margin of the participation decision are most sensitive to perceived costs, so interventions that increase complexity may disproportionately deter individuals closest to acting voluntarily. Finally, digital delivery introduces constraints beyond information design, including technological literacy and user experience, that can generate disengagement even before content is encountered.

More broadly, these results challenge the assumption that accurate and accessible information is at worst neutral in settings targeting vulnerable populations. In contexts involving complex administrative decisions, informational interventions can reduce participation by raising perceived costs among individuals who are already close to acting voluntarily. The barriers to take-up in such settings are therefore not purely informational, and interventions targeting those barriers must account for how information interacts with perceived complexity and individual heterogeneity in baseline motivation. Digital

RETURNS TO CAPITAL IN MICROENTERPRISES: EVIDENCE FROM A FIELD EXPERIMENT

Suresh de Mel, David McKenzie and Christopher Woodruff*

Abstract

Small and informal firms account for a large share of employment in developing countries. The rapid expansion of microfinance services is based on the belief that these firms have productive investment opportunities and can enjoy high returns to capital if given the opportunity. However, measuring the return to capital is complicated by unobserved factors such as entrepreneurial ability and demand shocks, which are likely to be correlated with capital stock. We use a randomized experiment to overcome this problem, and to measure the return to capital for the average microenterprise in our sample, regardless of whether or not they apply for credit. We accomplish this by providing cash and equipment grants to small firms in Sri Lanka, and measuring the increase in profits arising from this exogenous (positive) shock to capital stock. After controlling for possible spillover effects, we find the average real return to capital to be 5.7 percent per month, substantially higher than the market interest rate. We then examine the heterogeneity of treatment effects to explore whether missing credit markets or missing insurance markets are the most likely cause of the high returns. Returns are found to vary with entrepreneurial ability and with measures of other sources of cash within the household, but not to vary with risk aversion or uncertainty.

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1. Introduction

Small and informal firms are the source of employment for large sectors of the labor force in many developing countries. A key question of interest is whether these firms hold the potential for income growth for their owners, or whether they merely represent subsistence sources of incomes for low-productivity individuals unable to find alternative sources of work. Evidence that *some* firms have high marginal returns is seen in the very high interest rates paid by some individuals to moneylenders, and by literature which identifies the effect of credit shocks on those who apply for credit (see Banerjee and Duflo, 2005 for an excellent recent summary). However, the sample of firms who apply for credit or who belong to microfinance organizations may not represent the full universe of firms for any number of reasons.¹ In this paper we use a randomized experiment to identify the marginal return to capital for all firms, irrespective of whether or not they choose to apply for credit at market interest rates. We then examine the heterogeneity of returns in order to test which theories can explain why firms may have marginal returns well above the market interest rate.

We accomplish this by surveying microenterprises in Sri Lanka and providing small grants to a randomly selected subset of the sampled firms. The grants were either 10,000 Sri Lankan rupees (about \$100), or 20,000 rupees (about \$200). Half the grants were given in the form of cash, and the other half in the form of equipment or inventories selected by the entrepreneurs. We purposely restricted our survey to small firms with less than 100,000 rupees (about \$1000) in capital other than land and buildings. The median level of capital investment among firms in the sample is about 18,000 rupees. The capital shocks provided were therefore large relative to the size of the firms, with the larger shock equal to more than 100 percent of the median level of invested capital.

An accurate estimate of returns to capital in small enterprises is important for understanding both the limitations and the potential of microenterprises to play a role in increasing income levels in developing countries. If returns are very low at low levels of capital stock, then this would indicate that capital is an important barrier to opening a

¹ These reasons include both selection among entrepreneurs as to whether or not to apply for credit, determined by factors such as their attitudes to risk, access to alternative sources of finance, perceptions of the returns on investment, and expectations of getting a loan; as well as selection on the part of the lender as to which firms to accept as clients.

per month, much higher than the market interest rate, and cannot rule out linearity in returns.

We then set out a model consistent with our data which can be used to investigate the proximate causes of such high returns to capital. The two main theories we consider are imperfect credit markets and imperfect insurance markets. Examination of the heterogeneity of treatment effects leads us to find that returns to capital are higher for entrepreneurs with higher ability and with fewer other wage workers in the household. We do not find significant interaction effects consistent with risk aversion or uncertainty limiting investment in the enterprise. This suggests that the reason for high returns to capital is missing credit markets rather than missing insurance markets, and that the household internal capital market is a binding constraint.

We begin by describing the experiment. We follow in Section 3 with a description of the data. Section 4 estimates the treatment effect and Section 5 the return to capital. Sections 6 and 7 consider robustness to spillovers, attrition, reporting differences, different treatments, and nonlinearities. In Section 8, we develop a simple model of household investment which incorporates both the riskiness of investment and factors affecting the shadow value of capital. The model provides several testable implications, which we confront with the data. Section 9 concludes.

2. Description of Experiment

Selection of the participants

We carried out a baseline survey of microenterprises in April 2005 as the first wave of the Sri Lanka Microenterprise Survey (SLMS).³ Four additional waves of the panel survey were then conducted at quarterly intervals. The survey took place in three Southern and South-Western districts of Sri Lanka: Kalutara, Galle and Matara. The survey is designed to also study the process of recovery of microenterprises from the December 26, 2004 Indian Ocean tsunami, and so these districts were selected as ones where coastal areas had received some tsunami damage. We set out to draw a sample of firms with invested capital of 100,000 rupees (about US\$1000) or less, excluding

³ Fieldwork was carried out by ACNielsen Lanka (Pvt) Ltd.

investments in land and buildings. This size cut-off was chosen in order that our treatments (described below) would be a large shock to business capital.

We began by using the 2001 Sri Lankan Census to select 25 Grama Niladhari divisions (GNs) in these three districts. A GN is an administrative unit containing on average around 400 households. We used the Census to select GNs with a high percentage of own-account workers and modest education levels, since these were most likely to yield enterprises with invested capital below the threshold we had set. The GNs were stratified according to whether the area was directly affected, indirectly affected, or unaffected by the tsunami. A door-to-door screening survey was then carried out among households in each of the selected GNs. This survey was given to 3361 households, with less than 1 percent of households refusing to be listed. The screening survey identified self employed workers outside of agriculture, transportation, fishing, and professional services who were between the ages of 20 and 65 and had no paid employees.

The full survey was given to 659 enterprises meeting these criteria. After reviewing the baseline survey data, we eliminated 41 enterprises either because they exceeded the 100,000 rupee maximum size or because a follow-up visit could not verify the existence of an enterprise. The remaining 618 firms constitute the baseline sample. For the purposes of this paper we exclude the firms which were directly affected by the tsunami, since the recovery of assets damaged by the tsunami might affect returns to capital. This gives a baseline sample of 405 firms which were not directly affected by the tsunami. The remainder of the paper will focus only on these firms.

The 405 firms are almost evenly split across two broad industry categories, with 203 firms in retail sales and 202 in manufacturing/services. Firms in retail sales are typically small grocery stores. The manufacturing/services firms cover a range of common occupations of microenterprises in Sri Lanka, including sewing clothing, making lace products, making bamboo products, repairing bicycles, and making food products such as hoppers and string hoppers.

The Experiment

The aim of our experiment was to randomly provide firms with a positive shock to their capital stock, and measure the impact of this on business profits. Firms were told

before the initial survey that as compensation for participating in the survey, we would conduct a random prize drawing, with prizes of cash or equipment for the business. The prize consisted of one of four grants: 10,000 Sri Lankan rupees (~\$100) in equipment for their business, 20,000 rupees in equipment, 10,000 rupees in cash, or 20,000 rupees in cash. In the case of equipment grants, the equipment was selected by the enterprise owner, and purchased by research assistants working for the project.⁴

After the first round of the survey, 124 firms were randomly selected to receive a treatment, with 84 receiving a 10,000 treatment and 40 receiving a 20,000 treatment. The randomization was done within district (Kalutara, Galle, and Matara) and zone (unaffected and indirectly affected by the tsunami). A second lottery was held after the third round of the survey, with an additional 104 firms selected at random from among those who didn't receive treatment after the first round: 62 receiving the 10,000 treatment and 42 the 20,000 treatment. In each case half the firms receiving a treatment amount received cash, and the other half equipment. Allocation to treatment was done *ex ante*, and as a result, there were an additional seven firms who were assigned to receive treatment after round 3, but who had attrited from the survey by then.

The 10,000 rupee treatment was equivalent to about three months of median profits reported by the firms in the sample, and the larger treatment equivalent to six months of median profits. The median initial level of invested capital, excluding land and buildings, was about 18,000 rupees, implying the small and large treatments correspond to approximately 55 percent and 110 percent of the median initial invested capital. By either measure, the treatment amounts were large relative to the size of the firms.

Although the amount offered for the equipment treatment was either 10,000 rupees or 20,000 rupees, in practice the amount spent on inventories and equipment sometimes differed from this amount. Among those receiving equipment, only a handful of firms spent less than the amount we offered. Only four of the 116 firms receiving equipment treatments spent as much as 50 rupees (\$0.50) less than the amount we

⁴ In order to purchase the equipment for these entrepreneurs receiving equipment treatments, research assistants visited several firms in the evening to inform them they had won an equipment prize. The winning entrepreneurs were asked what they wanted to buy with the money, and where they would purchase it. The research assistants then arranged to meet them at the market where the goods were to be purchased at a specified time the next day. Thus, the goods purchased and the place/market where they were purchased were chosen by the entrepreneurs with no input from the research assistants.

have worked in creating groups which are comparable in terms of baseline characteristics. There is no significant difference in the baseline profits, revenue, capital stock, age, and level of informality of the treated and untreated firms. The age, education, parental background, and risk aversion are similar for owners in both groups. The only significant difference between the two groups occurs for a household durable asset index, with firm owners in the control group having slightly higher mean baseline assets. To test for joint significance of all these variables, we estimate a probit model of whether or not a firm is assigned to treatment as a function of the 16 baseline characteristics given in the Table. The p-value from the chi-squared test is 0.355, so that we can not reject the null hypothesis that these observable characteristics do not help predict whether or not a firm was assigned to treatment.

3. Data/Survey

The baseline survey gathered detail information on the firm and the characteristics of the firm owner. The main outcome variable of interest in this paper is the profits of the firm. Firm profits were elicited directly from the firm by asking:

“What was the total income the business earned during March after paying all expenses including wages of employees, but not including any income you paid yourself. That is, what were the profits of your business during March?”

The reported mean and median profits in the baseline are 3900 rupees and 3000 rupees respectively. The survey also asked detailed questions on revenues and expenses. Profits calculated as reported revenues minus reported expenses are lower, around 2500 at the mean and 1350 at the median. Profits calculated in this manner are positively correlated with reported profits, with a correlation coefficient of 0.32. This is about the same level as one finds in other microenterprise surveys. In de Mel, McKenzie and Woodruff (2007), we analyze the measurement of profits in some detail, reporting on experiments conducted with different questions, bookkeeping, and monitoring of sales. The biggest reason reported and calculated profits differ is a mismatch of the timing of purchases and the sales associated with those purchases. Some of the expenses in one

month are associated with sales the following month. Correcting for this mistiming increases the correlation with reported profits to around 0.70. We conclude from the more detailed analysis of measurement issues that the reported profit is the best measure of the firm's profitability, and we use those data for the remainder of this paper.

The baseline survey also gathered detailed information on the replacement cost of assets used in the enterprise, and whether they were owned or rented. 99 percent of assets excluding land and buildings are owned by the enterprises. The majority of assets owned by the enterprises are land and buildings. In the baseline sample, these average 121,000 rupees (\$1200), though about a sixth (15 percent) of firms report they own no assets in this category. The firms also reported an average of 14,400 (\$145) rupees worth of machinery and equipment and 13,000 (\$130) in inventories.

In each subsequent round of the survey, we then asked firms to report on the purchase of new assets, the disposition of assets by sale or damage, and the repair and return to service of assets. These are used to calculate a change in assets over the quarter. Combined with the data from each previous quarter, these data allow us to estimate investment levels by category for each quarter of the survey. Note that the data are not adjusted for depreciation of machinery and equipment. Given the relatively short period over which the data are gathered and the high percentage of assets invested in inventories, we do not see this as a major concern. The survey also asks the current value of inventories of raw material, work in progress and final goods each quarter.

We began with 405 firms in the baseline survey, and still had 384 of the firms in the survey by the fifth wave, an attrition rate of only 5.2 percent. However, only 388 of the 405 firms reported profits in the first wave, 390 in the second wave, 381 in the third, 366 in the fourth, and 366 in the fifth. We concentrate our analysis on the unbalanced panel of 383 firms reporting at least three waves of profit data. Fifteen of the firms assigned to the control group and 7 of the firms assigned to the treatment group attrited from the sample before reporting at least three waves of profit data. We show our results are robust to this attrition.

4. Estimation of the Mean Treatment Effect

We begin by assuming each enterprise is characterized by a linear production function, and that the treatments have homogeneous effects on the enterprises. We later relax both of these assumptions. Estimating the mean effect of treatment on business profits requires estimating for firm i in period t :

$$PROFITS_{i,t} = \alpha + \beta AMOUNT_{i,t} + \sum_{s=2}^5 \delta_s + \varepsilon_{i,t} \quad (1)$$

Where $AMOUNT_{i,t}$ is an indicator of how much treatment firm i had received at time t , coded in terms of 10,000 rupees. $AMOUNT$ therefore takes value 1 if the firm received the 10,000 rupee treatment, 2 if the firm received the 20,000 rupee treatment, and 0 otherwise. The δ_s are period effects. Table 1 showed that the randomization appears to have worked, giving treatment and control groups which are comparable in terms of observable baseline characteristics. Hence the expected correlation between the error term $\varepsilon_{i,t}$ and treatment status is zero. We can therefore estimate equation (1) using random effects, which assumes $\varepsilon_{i,t}$ to be uncorrelated with treatment status, but allows it to be correlated over time. Alternatively, we can use fixed-effects estimation, which will improve the precision of the estimates by capturing any time-invariant characteristics of the firm which affect business profits.

The first row of Table 2 presents the resulting treatment effects on nominal profits. Column 1 shows the random effects estimate and Column 2 the fixed effects estimate. The coefficients show a 10,000 rupee treatment increased monthly profits by 489 to 569 rupees, a monthly nominal return on the treatment of 4.9 to 5.7 percent, with these treatment effects significant at the 5 percent level. The profits data are noisy, and so Columns 3 and 4 present results after trimming outliers. In Column 3 we trim firms with monthly nominal profits below 200 rupees or above 25,000 rupees, corresponding to the 1st and 99th percentiles of the nominal profits distribution. This increased the estimated nominal return on the treatment to 6.1 percent, now significant at the 1 percent level. A firm that always has profits above 25,000 rupees is likely to just be a very productive firm, whereas a firm whose profits change from 1500 to 30,000 to 2000 could be one in which misreporting occurred. Column 4 therefore presents our preferred specification, which trims firms on the basis of extremely large *changes* in profits. We trim firms where

the change in monthly profits from one quarter to the next was less than -90 percent or greater than 962 percent, corresponding to the 1st and 99th percentiles of the distribution of changes in profits. The estimated nominal return on the treatment is 5.9 percent per month after this trimming.

Inflation in Sri Lanka was low over the period studied, with point-to-point inflation of 4.0 percent between March 2005 and March 2006. The second row of Table 2 examines the treatment effect on real profits, where the monthly Sri Lanka Consumers' Price Index was used to deflate nominal profits.⁶ After trimming based on extreme changes in profits, we see that the real monthly increase in profits was 5.7 percent of the treatment amount.

Impact on Labor Hours

This treatment effect represents the increase in real monthly profits from our treatment. The treatment increased capital stock. However, labor hours may have also have been affected if firm owners adjusted their own labor hours, or if the resulting rise in profits caused owners to substitute towards leisure. We therefore empirically investigate the impact of the treatment on labor hours by estimating equation (1), using own labor hours and family worker labor hours as the dependent variables.⁷ The treatment is found to have a positive and significant impact on own labor hours: a 10,000 rupee treatment increases own weekly hours by 3.4 hours ($t=3.23$, $p=0.001$). There is a small and insignificant negative impact on family labor hours (a -0.7 hour per week effect, $p=0.467$).

Recall that business profits include the earnings of the firm owner. Hence the increase in real profits from the treatment reflects both the return to the additional capital, and the return to the owner of the additional labor hours worked. To isolate the impact of capital, we need to subtract the implicit wage earned by the firm owner. We estimate the marginal return to own labor by using the baseline data to regress profits on capital stock (exclusive of land and buildings), age of the owner, owner's education, a dummy for the

⁶ Source: Sri Lanka Department of Census and Statistics, http://www.statistics.gov.lk/price/slcpi/slcpi_monthly.htm [accessed February 17, 2007].

⁷ Recall that none of the firms had paid non-family workers in the baseline, and less than 5 percent of firms had non-family workers at any time during the panel survey.

$$PROFITS_{i,t} = \alpha + \beta K_{i,t} + \sum_{s=2}^5 \delta_s + \varepsilon_{i,t} \quad (2)$$

where the treatment amount offered to firms is used to instrument capital stock, $K_{i,t}$.

Table 3 reports the estimated return to capital from estimating equation (2). Columns 1 through 4 repeat the specifications used in Table 2, using random effects, fixed effects, trimming the level of profits, and trimming changes in profits. Columns 5 through 8 adjust for potential outliers in capital stock, by additionally trimming the top and bottom 1 percent of initial capital stocks. After trimming profits we get real returns to capital in the 6.8-6.9 percent per month range, which increase to 7.3-7.4 percent once we trim outliers in initial capital stock.

The lower panel of Table 3 shows the coefficient on the treatment amount in the first stage regression for capital stock. The treatment amount is a strongly significant predictor of changes in capital stock (F-statistic always above 41), with each 1000 rupees of treatment increasing capital stock by 744-800 rupees. This explains why the return on capital is higher than the treatment effect, since the change in capital stock is less than the treatment amount.

6. Controlling for potential Treatment Spillovers

The key condition for randomization to provide us with a valid estimate of the treatment effect and of returns to capital is the standard unit treatment value assumption, which requires that the potential outcomes for each firm are independent of its treatment status, and of the treatment status of any other firm (Angrist, Imbens and Rubin 1996). As Miguel and Kremer (2004) and Duflo, Glennerster and Kremer (2006) note, the presence of spillovers can cause this assumption to be violated, leading to biased estimates of the treatment effect. It is therefore important to test for spillover effects arising from our grants to firms.

We collected the GPS coordinates of each firm in our survey, taking advantage of improvements in precision and technology which allow location to be measured accurately to within 15 meters, 95 percent of the time (Gibson and McKenzie, 2007). This allows us to construct a measure of the number of treated firms in the same industry at any given point in time within a given radius of each firm. In our baseline survey, the

7. Further Robustness

Robustness to Attrition

The attrition rate in the SLMS was remarkably low for a five-wave panel survey. Nevertheless, the combination of attrition, missing profit data, trimming capital stock, and trimming firms with large changes in profits reduces our sample of non-bamboo firms from 376 to 325 firms. Of the 51 firms which are dropped, 24 were assigned to treatment and 27 to the control group. This corresponds to a broad attrition rate (including both attritors and those dropped for missing or noisy data) of 11.1 percent for the treated, and 16.9 percent for the control group. To examine the robustness of our results to this differential attrition, we use the bounding approach of Lee (2005) to construct upper and lower bounds for the treatment effect.

The key identifying assumption required for implementing the Lee (2005) bounds is a monotonicity assumption which assumes that treatment assignment affects sample selection only in one direction. In our context, it requires assuming that there are some firms who would have attrited if they had not been assigned to treatment, but that firms do not attrit because of being assigned to treatment. This seems plausible in our context, since firms receiving treatment may stay in the sample when they would otherwise have attrited as a result of a business failure being prevented, or to an increased willingness to answer our survey question. It does not appear likely that receiving treatment would have caused some firms to drop out of the survey that would not have dropped out if they had remained in the control group.⁹

To construct the Lee (2005) bounds we then trim the distribution of profits for the group assigned to treatment by the difference in attrition rates between the two groups as a proportion of the retention rate of the group assigned to treatment. In our application, this requires trimming the upper or lower 6.5 percent of the adjusted profits distribution for the group assigned to treatment. Doing this then gives a lower bound for the treatment effect of 348 rupees and an upper bound of 399 rupees. The returns regressions

⁹ Of course attrition from our estimation sample also arises from an outlier change in profits. Conceivably the treatment might cause a large increase in profits, making attrition for this reason more likely. However, it would have to increase profits by more than 962 percent in order to do this, and if this were the case, dropping such observations will only lower the estimated treatment effect.

Together these results strongly suggest that the estimated returns are indeed true returns, and not merely a consequence of firms changing reporting of profits in response to treatment. We now turn to examining how the returns vary by type of treatment.

Cash vs. Equipment Treatments and Linearity of Returns

The analysis thus far has pooled together the treatments provided as cash and equipment, and the 10,000 and 20,000 rupee treatment levels. Table 6 examines whether the impact of the equipment treatment varies from that of the cash treatment. The first column repeats the overall treatment effect in row 3, Column 4 of Table 2. Column 2 then separates the treatment into two variables, one representing the equipment treatment amount and the other representing the cash treatment amount. We see that the coefficients on the equipment and cash treatments are nearly identical, 370 on equipment and 393 on cash. Since we can not reject equality of the two treatment effects, we will continue to pool the two treatments in the remaining analysis.

The third column of Table 6 uses two separate dummy variables to examine the impact of the 10,000 and 20,000 rupee treatments, with the sample excluding firms in the bamboo industry and trimmed at the 1st and 99th percentile of changes in profits. The 10,000 treatment is estimated to raise net profits by 480 per month, compared to a 721 increase for the 20,000 treatment. However, the standard errors in both point estimates are high. We can reject neither the hypothesis that the effect of the 20,000 treatment is twice that of the 10,000 treatment, nor the hypothesis that the two treatments have the same effect.

Column 4 of Table 6 provides an alternative test of non-linearity of treatment effects by interacting the amount of the treatment with the baseline level of capital stock, exclusive of land and buildings. We also interact the period effects with this initial level of capital stock to allow for different growth paths of firms according to their initial capital stock levels. Column 4 shows the results after 1 percent trimming of firms on the basis of changes in profits. The interaction term is negative, which would suggest that the return to capital is decreasing, but is not statistically significant.

As a further check as to whether the treatment effect varies with firm size, we use propensity score matching to match each treated firm which received the 10,000 rupee

93 percent of firms say is a constraint. The second most prevalent constraint, lack of inputs, which 53 percent of firms list as constraint, is also likely to reflect in part liquidity constraints, as firms said that they couldn't afford to buy all the inputs they would need. The perception of financial constraints is supported by the relatively rare use by firms of formal finance. Only 3.1 percent of our firms have a bank account for business use, and 89 percent of firms got *no* start-up financing from a bank or microfinance organization. Formal credit is scarcely used at all for financing additional equipment purchases. Instead the major source of both start-up and ongoing funds is personal savings of the entrepreneur and loans from family. On average 69 percent of start-up funds came from this source, and 71 percent of firms relied entirely on own savings and family for start-up funds.

After finance, the second most common set of constraints to growth according to the firms themselves can be broadly interpreted as reflecting uncertainty among firms about realizing the gains from investment. The possibility of lack of demand for products (which 34 percent of firms say is a constraint), lack of market information (16 percent say is a constraint), and economic policy uncertainty (15 percent say is a constraint) all suggest that the riskiness of returns could be important. Relatively few firms view the legal and regulatory framework as constraints to growth: 16 percent say lack of clear ownership of land, 6 percent say legal regulations, and only 1 percent say high tax rates is a constraint. Few firms view infrastructure as a constraint.

These perceptions suggest that missing markets for credit or for insurance against risk could be important factors in explaining the high marginal returns to capital. These are two of the most common explanations considered in the literature.¹⁴ We provide a simple model of microenterprise production to illustrate how these missing markets can give rise to marginal returns in excess of the market interest rate, and which can be used along with evidence of the heterogeneity of returns from our treatment to test whether missing credit or insurance markets is the more important factor driving high returns.

¹⁴ See Banerjee and Duflo (2005) for an excellent recent review of different explanations. Missing credit and insurance markets appear the most important for our setting among the different theories they summarize.

returns. The median digit span recall is 6 digits, corresponding to an estimated return of 4.8 percent. At the top of the distribution, 11 percent of firm owners could recall 8 or more digits, resulting in predicted returns of 13.6 percent per month.

Column 8 shows no significant interaction of the treatment amount with risk aversion. Moreover, the sign of the coefficient is negative, suggesting that, if anything, firm owners which are more risk averse have lower returns. Column 9 finds a negative coefficient on uncertainty of firm profits, which is significant at the 10 percent level. This would suggest firm owners facing more uncertainty have lower returns. This is evidence against the high marginal returns being caused by risk averse entrepreneurs operating with missing insurance markets, as this would predict that both coefficients would be significantly positive.

Finally in Column 10 we consider the interactions together. The results are very similar to the regressions examining each effect individually. The interactions with the number of paid wage workers in the household and education are still positive and significant at the 5 percent level, confirming that the returns are highest for owners with more limited liquidity and more ability. Risk aversion is insignificant, and the measure of sales uncertainty has the unexpected sign and is significant at the 10 percent level.

Taken together, the heterogeneity of returns supports the view that the high marginal returns from our treatments reflect credit constraints rather than missing insurance markets. Credit constraints are more binding, and thus marginal returns are higher, for more able entrepreneurs and for entrepreneurs with a high shadow cost of capital within the household, measured by the presence of fewer other paid wage workers.

9. Conclusions:

We find returns to capital for the average microenterprise firm in our sample to be very high, around 5.7 percent per month, or at least 68 percent per year. Returns appear to be flat or decreasing – we do not find evidence of increasing returns over our sample range. Marginal returns are highest for entrepreneurs with more ability and with fewer other workers in the household. These are the individuals who, all else equal, we would expect to be most credit constrained. In contrast, returns are not found to differ with risk

aversion of the entrepreneur, or with the perceived uncertainty of the entrepreneur about his or her profits. This suggests that missing credit markets rather than missing insurance markets is the main constraint on enterprise growth.

The results have important implications for the way we think about both microenterprises and microfinance markets. The high returns at very low levels of capital stock imply that non-convex production sets are unlikely to lead to permanent poverty traps. The high returns also suggest that there will be demand for loans from microlenders, even at interest rates which are high enough to cover the administrative costs of making small loans.

The fact that the returns exceed interest rates charged by microlenders in Sri Lanka by such a large margin also presents something of a puzzle. Sri Lanka has an active microfinance sector, with participation by both NGOs and government-run banks. Initial loan sizes are in the 10,000 rupee range, the same as our smaller treatment level. Do these enterprises lack credit because they have been screened out, in spite of the high returns? Or do they lack credit because of a lack of demand, either because they are unaware of potential sources of credit or because they are unaware of the returns to incremental investments in the enterprises? These questions are left as motivation for future research.